

Technical Working Group Recommendations for Catch Record Card-based Puget Sound Recreational Crab Harvest Estimation

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GLOSSARY

1. Arithmetic Mean: A type of average of a group of numbers, calculated by summing all numbers and dividing this sum by the number of numbers in the group.
2. Bias Correction Factor (Non-Response): A factor generated by the NRBCM to offset the observed difference in catch between the population of CRC-holders that report their catch and the population of CRC-holders that do not report their catch.
3. Catch: Dungeness crabs retained by recreational crabbers, with or without a catch record card (CRC).
4. Catch Estimate: A summation of crab caught in the Puget Sound recreational fishery which includes catch reported directly on CRCs and to WDFW as well as estimates of catch retained but not recorded or reported on CRCs to WDFW.
5. Catch per Card Method: A method for estimating non-respondent catch using the mean percentage of successful non-respondents (from the 2022 and 2023 Responsive Management surveys) and the mean catch per card of the successful, on-time respondents for the season.
6. Catch per Card: The total number of Dungeness crab recorded on a CRC.
7. Catch Record Card (CRC): Catch Record Card. A mandatory catch reporting tool for recreational crabbers in Puget Sound. CRCs are issued for summer and winter seasons and must be submitted by a set deadline, either online or by mail, to avoid penalties.
8. Census: Catch on CRCs that is directly reported to WDFW (on-time or late), not including estimates for unreported catch.
9. Charter Catch: Catch of Dungeness crab in the Puget Sound recreational fishery that is harvested as part of a licensed charter operation and reported on a charter stamp submitted directly to the WDFW by the charter license holder, not the angler.
10. Co-manager: A Tribal, state, or federal government engaged in shared government to government consultation and management of a resource.
11. CRC Batch: A set of mailed-in CRCs that are organized based on the date they are received and batch coded based on when they were post-marked. If a post-mark is not discernible then the post-mark date is assumed based on the post-mark dates of mail received at the same time. Typically, the time interval for grouping is daily (within reporting period) to weekly (after reporting period).

12. CRC Database: A central repository for collocating catch reporting data from the various reporting sources which allows WDFW to process, perform analysis and generate estimates from a standardized data set.
13. CRC Reporting Deadline(s) (on-time): The date before which CRC's are considered to be on-time. The on-time reporting deadlines are October 1 for summer CRC and February 1 for winter CRC.
14. CRC-holders: Individuals with CRCs issued in their name.
15. Creel: An in-person survey in which an interviewer asks a harvester about their fishing experience and collects information about the catch and trip details.
16. Direct-Report Catch per Trip: A method for estimating non-respondent catch using the mean number of trips by successful non-respondents reading from their cards (from the 2022 and 2023 Responsive Management surveys) and the mean catch per trip of the successful, on-time respondents for the season.
17. Estimated Unreported Catch (EUC) (also sometimes referred to as Estimated Unrecorded Catch): The catch that is retained but not reported on a CRC, both by CRC holders and by non-CRC holders.
18. Final Recreational Catch Estimate: A summation of crab caught in the Puget Sound recreational fishery which includes catch reported directly on on-time CRCs as well as directly reported late CRC catch, estimates of catch retained but not reported on CRCs by non-respondents, and Estimated Unreported Catch.
19. Late Card Submission Cut-off Date: the date by which CRC's must be received to be included as part of the CRC- census catch estimate. For the 2025, the submission date was agreed to be May 15 for both summer and winter CRCs.
20. Late Catch Record Card: A CRC turned in after the seasonal reporting deadline but before the late CRC reporting cut-off date.
21. Marine Area: "Marine Areas" are the State waters of Washington, within Puget Sound including Hood Canal, the Strait of Juan de Fuca, the San Juan Islands, and the Strait of Georgia. Marine Areas are the spatial management unit at which recreational crab rules, regulations, and harvest are set for the estimates made in this document. Marine Area boundaries are described in WAC 220-311-010.

22. Model-averaging Approach: A method for estimating non-respondent catch by taking the average of the 'Catch per Card' and 'Direct Report Catch per Trip' catch estimation methods.
23. Non-respondent Catch: Dungeness crab retained by CRC-holders who do not return their CRC by the late CRC reporting cut-off date.
24. Non-respondent Surveys: Online or phone surveys of non-respondent CRC-holders to acquire catch data.
25. Non-respondent: A CRC holder who does not turn in the CRC data to WDFW by late CRC reporting cut-off date.
26. Non-response Bias Correction Model (NRBCM): A quadratic regression model used to predict the bias correction needed based on the reporting rate (i.e., proportion of CRCs returned).
27. Non-response Bias: The bias that occurs when respondents do not accurately represent the full population being sampled; in fisheries, non-respondents tend to catch less than respondents.
28. Northwest Indian Fisheries Commission (NWIFC): The organization that supports fishery fishery management for treaty Tribes in western Washington.
29. On-time Respondent: A CRC holder who turns in their CRC data to WDFW by one of the following means: online via WILD or email, by phone, by mail, or by drop-off at a WDFW office before the official seasonal-specific on-time reporting deadline.
30. Preliminary Catch Estimate: A catch estimate that is based on catch reported on all CRC's (both internet and paper submissions) returned by the on-time CRC reporting cut-off date(s). The preliminary catch estimate also includes a ratio estimate of late-reported catch, as well as non-respondent catch, and Estimated Unreported Catch.
31. Ratio-based Model: Model using calculated, proportional relationships between data points to analyze, forecast, and benchmark performance.
32. Recall Bias: Decreased accuracy in self-reporting of past catch from memory.
33. Recreational Crabber: An individual fishing in the state of Washington's recreational Dungeness crab fishery for personal use, with a CRC card or not.
34. Reporting Rate: The percent of CRCs returned out of CRCs issued.
35. Respondent: A CRC holder who returns their card and whose catch is directly censused.

36. Responsive Management Surveys: Phone and online surveys of non-respondent CRC-holders performed in 2022 and 2023 conducted by the natural resources survey firm, [Responsive Management](#).
37. Retained Crab: Dungeness crab that is caught and not released by a recreational crabber, including both reported and unreported crab.
38. Season: The period of year in which a recreational fishery occurred. Two separate crab seasons are offered in the recreational fishery, summer and winter. The summer season occurs early July through labor-day, with the exception of marine areas 7N and 7S which start mid-July and mid-August, respectively. The winter season occurs October 1 through December 31.
39. Shellfish Implementation Plan (SIP): A court-approved order that governs how the Washington treaty Tribes, the State of Washington, the United States, and private tideland owners will implement the 1994 “Rafeedie Decision”, a sub-proceeding of US v. Washington. The SIP provides a framework, principles and course of action for effective cooperative management of shellfish resources in Washington State.
40. Successful Crabber: A crabber who has retained at least one Dungeness crab in a trip or season.
41. Successful Trip: A trip where at least one Dungeness crab is retained.
42. Summer CRC: A catch record card valid for recording recreational Dungeness crab catch from Puget Sound June 1 through Labor Day.
43. Summer Fishery: The Puget Sound recreational Dungeness crab fishery occurring early July through labor-day, with the exception of marine areas 7N and 7S which start mid-July and mid-August, respectively.
44. Technical Review Panel (TRP): A court adjudicated panel of experts that reviewed the methodology for estimating unreported catch in the Puget Sound recreational fishery and provided recommendations on ways to improve the estimate for unreported catch.
45. Technical Working Group (TWG): Joint Shellfish Technical Workgroup, consisting of shellfish biologists, harvest managers and biometricians from the Washington Department of Fish and Wildlife (WDFW), participating treaty Tribes, and Northwest Indian Fisheries Commission (NWIFC).
46. Total Summer Catch: All crab retained by all harvesters in the summer recreational fishery (reported and unreported).

- 47. Trip: An individual crabbing event, successful or unsuccessful, that contributes to overall daily catch.
- 48. Washington Department of Fish and Wildlife (WDFW): The state government agency in Washington charged with managing fish and wildlife resources.
- 49. Winter CRC: A catch record card valid for recording recreational Dungeness crab catch from Puget Sound the day after Labor Day through December 31st.
- 50. Winter Fishery: The Puget Sound recreational Dungeness crab fishery occurring after Labor Day through December 31st.

BACKGROUND/CURRENT FRAMEWORK

The reporting of catch associated with the State of Washington's Puget Sound recreational Dungeness crab fishery has been a long-standing concern for tribal shellfish co-managers. The Puget Sound Dungeness crab fishery's recreational *Catch Record Card* (CRC) catch accounting system has been a particularly contentious issue among state and tribal shellfish managers. As a result, the shellfish Technical Working Group (TWG) was formed to address this issue per Section 5 of the Shellfish Implementation Plan (898 F. Supp. 1453 [1994]). The Washington Department of Fish and Wildlife (WDFW) and affected Puget Sound tribal co-managers have expressed valid concerns regarding recreational catch estimation methods which include:

- a generally less than 50% compliance rate for returned CRC with a reduction in CRC reporting rates since 2019;
- a shift from mail in CRCs to on-line reporting of CRCs;
- lack of transparency associated with CRCs and the CRC catch estimate methods;
- use of post-season phone and in-season creel surveys to derive parameters of self-reporting CRC-holders;
- outdated data to inform the estimated unreported catch (EUC) proportion and
- outdated data to inform the non-response bias correction model (NRBCM)

In 2023, WDFW reported to tribal co-managers a series of issues with the CRC data processing procedures that were implemented to generate recreational catch estimates. Paramount among these was the incorrect and inconsistent application of the non-response bias correction among years. Ultimately, these errors resulted in recreational catch estimates that differed from what was previously communicated. Limited sharing of data with tribal co-managers and incomplete documentation of data processing methods prevented reliable reconstruction of historical recreational harvest estimates. These issues underscored a lack of shared understanding of the State's Puget Sound recreational crab catch estimation methods, highlighting the need for greater transparency in data sharing and data processing documentation.

In response to the specific issues above, as well as long-standing concerns expressed by tribal co-managers with upholding the 50:50 sharing principle (Shellfish Implementation Plan, 898 F. Supp. 1453 [1994]), staff from the Northwest Indian Fisheries Commission (NWIFC), WDFW, and the treaty Tribes initiated the TWG in 2024 to objectively and collaboratively address issues impacting shellfish harvest management. With WDFW and Tribal policy direction, the TWG was initially charged with building a shared understanding of recreational catch estimation and data processing procedures and to develop sound technical methods for the CRC-based recreational crab catch estimate. State and tribal managers engaged in lengthy technical discussions during the summer of 2024 and through 2025. By the end of August 2025, many of the data processing procedures needed for reconstruction of the 2025 recreational CRC catch estimates were addressed and TWG recommendations and this report were prepared.

The TWG's initial focus was on on-time and late reporting, and initial recommendations by the TWG were presented to co-manager policy in fall of 2024 (see Appendix 2). As shown in Figure 1, harvest from *on-time* respondents – those who submitted Catch Record Cards (CRC) to WDFW by the on-time CRC reporting deadline either by mail or online – comprises the largest portion of the recreational catch estimate. While many of the 2024 TWG recommendations have already been implemented, several general recommendations and items specific to on-time respondents remain outstanding. These outstanding items are described in the November 7, 2025, policy briefing and are currently awaiting a policy decision or implementation. A full account of these recommendations is included in Appendix 3.

The focus of this document is to provide discrete recommendations by the TWG on the Puget Sound recreational CRC-based catch estimate and data processing methods for those portions of the catch that are not directly reported to WDFW on CRCs:

- *LATE CATCH RECORD CARDS* – Evaluating and updating the recording and incorporation of catch data submitted to WDFW after the CRC submission deadline directly into the catch estimate.

- *CATCH RECORD CARD NON-RESPONDENTS* – Validating, documenting, and updating the methods used to estimate catch and effort by those recreational card holders that obtain a CRC but do not return it to WDFW.
- *ESTIMATED UNREPORTED CATCH (EUC)* – Evaluating the data available to inform the portion of catch not reported in the recreational crab fishery and establishing recommended procedures for regularly updating the EUC proportion using the methods described in the court-adjudicated Technical Review Panel (TRP) report (Hankin *et al.* 2021).

This technical report is a comprehensive companion document to the November 2025 policy briefing (Appendix 3). The purpose of this report is to provide the necessary background information regarding recreational catch estimation methods, and detail the recommendations made by the TWG specific to each topic for WDFW and tribal co-manager policymakers. Full documentation of all methods employed in the Puget Sound recreational crab catch estimation are described in the WDFW Technical Report (Bosley *et al.* in prep).

The current CRC-based framework has been used by WDFW to estimate recreational Dungeness crab catch in Puget Sound since 2007, as a means to overcome logistical and precision challenges associated with field-based catch assessment methods (i.e. vessel-based buoy surveys). In contrast to the methods available at the time of implementation, CRC-based estimates provided an efficient and cost-effective means of estimating catch across broad spatial areas at a high degree of temporal resolution. CRC-based estimates are not unique to the Puget Sound crab fishery. They are also used either as principal or backup estimates for catch reporting in other fisheries including salmon, steelhead, and sturgeon within Washington state, rock scallops in Oregon, and Spiny lobster in California. Participants in the Puget Sound recreational Dungeness crab fishery are required by law (WAC [220-310-010](#)) to submit CRCs detailing their catch by date and Marine Area and, if no catch occurred, submit a CRC with no catch recorded.

In most years since 2011, the framework for estimating recreational crab catch relied on a non-response bias correction model (NRBCM) that used catch from reported CRCs and application of a bias correction factor to account for differences in harvest between those who reported catch on their CRCs (“respondents”) and those who did not (“non-respondents”). The difference in average catch between respondents and non-respondents is known as “non-response bias”. Recreational catch is corrected for non-response bias due to observed differences in behavior between those who return their CRC’s and those who fail to return their CRC’s (Tarrant *et al.* 1993). Shortly after its implementation, tribal co-managers expressed concerns that harvest that was unreported was not being accounted for by the NRBCM.

Following multiple years of tribal policy concerns and the interim application of different Estimated Unreported Catch (EUC) values ranging from 4% to 10%, WDFW managers began applying a 6% add-on to the NRBCM-based recreational catch estimate to account for EUC which was incorporated into annual crab management plans at that time. The 6% add-on for EUC accounts for catch by individuals who do not hold a CRC and for catch by CRC-holders who fail to record all or some of their harvest. The 6% EUC value has since been formally disputed by tribal co-managers and was revised following a court-adjudicated Technical Review Panel’s (TRP) independent expert review of available Puget Sound-wide, creel data that spanned from 2010 to 2016 (Hankin *et al.* 2021). State managers began using a new rate of 9.97% for the Puget Sound-wide EUC in 2021 following the TRP’s recommendation.

From 2011 – 2022, annual recreational catch was intended to be estimated by expanding the reported catch by the proportion of issued cards to reported cards, then applying the non-response bias correction factor derived from the NRBCM (see 2011 – 2022 WDFW CRC memos for details). However, the errors in catch estimates identified by WDFW managers in 2023 revealed discrepancies in how the CRC data was being processed. It was determined that the non-response bias correction was not being applied

correctly or consistently across seasons, and there was a lack of documentation to inform which procedures were used in which years. The uncovering of these errors followed the 2021 TRP report which highlighted that: 1) the outdated 2007-2010 non-respondent telephone survey data utilized in the NRBCM was the “primary source of uncertainty” in the recreational catch estimates, and 2) the 2010-2016 creel data was unsuitable to inform population parameters of CRC-holders due to survey design limitations. Collectively, these issues led WDFW managers to collect updated non-respondent survey data in 2022 and 2023; design and implement a region-focused, probability-based creel survey of select access points; and improve and update CRC data processing rules and code (WDFW Technical Report: Bosley *et al.* in prep).

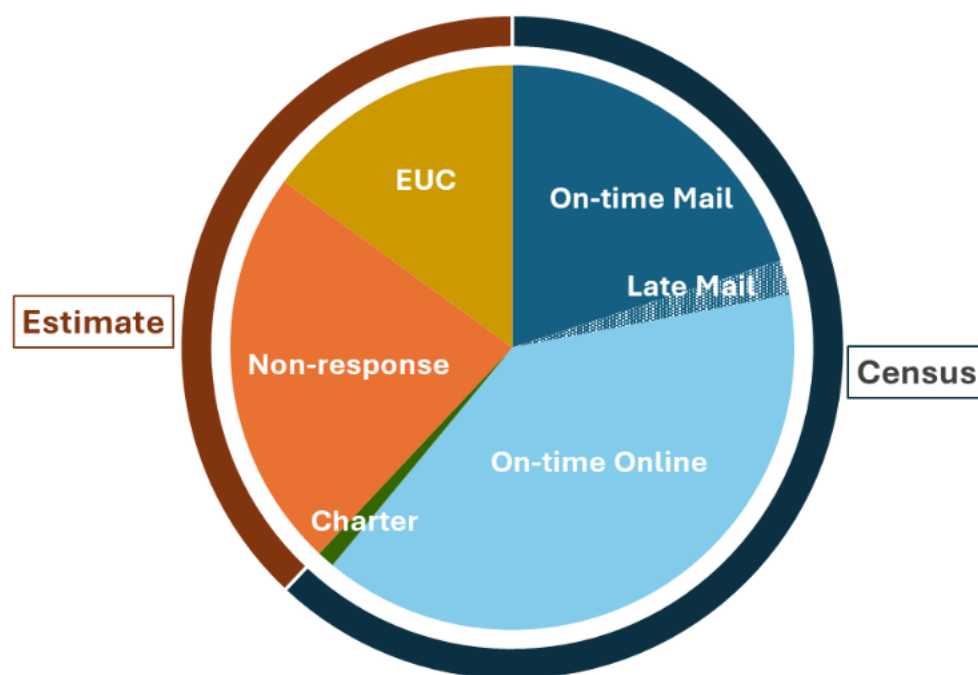


Figure 1 – Contributions by component of the recreational crab catch estimate. A pie chart representing each component contributing to the Puget Sound recreational Dungeness crab catch estimate. Pie wedges are not drawn to scale, as some portions are unknown and vary from year to year. The components of the catch estimate that can be counted directly from CRCs are identified as census, and the components that cannot be counted directly (EUC and non-respondent catch) must be estimated.

Through the formation of the TWG, there has been an improved understanding by co-managers of several components of the state recreational crab catch estimate. Since the summer of 2024, the TWG has met regularly to discuss the components of the CRC-based recreational crab catch estimate and develop recommendations for policy consideration. The TWG has strived to be objective in its assessment and evaluation to develop recommendations in good faith of the policy direction, and to make recommendations based on the consensus of its participants, when possible. In instances where consensus regarding a recommendation was not achieved, the TWG has either offered multiple recommendations in a non-prioritized order for policy to consider or the party(ies) not in agreement may provide recommendations in separate documents. Additionally, TWG members recognize that while some recommendations may represent the highest standard that could be implemented, policymakers may want additional options to consider, given the potential staff capacity, fiscal, or technical limitations that would need to be addressed to implement them.

LATE CATCH RECORD CARDS

Description

Each year, some state recreational crabbers submit their Catch Record Cards (CRCs) to WDFW after the on-time CRC reporting deadline(s) (an average of ~3% of summer CRCs and ~2% of Winter CRCs; 2014-2024). WDFW documents the receipt of late CRCs in order to impose a penalty fee, but the catch data on late CRCs was not historically entered into a central catch database. From 2007 to 2023, late cards were treated as non-respondents and catch data from late CRCs were not partitioned out separately, primarily due to the need to generate more timely final harvest estimates and due to staff capacity limitations (R. K. Childers, WDFW personal comm.). Prior to 2024, late CRCs were destroyed, therefore catch data from these CRCs are not available to be retroactively entered or analyzed. Based on 2024 recommendations from the TWG, late CRCs will no longer be destroyed by WDFW so that data can be retained, recorded, and analyzed. Under comanager policy direction, the TWG identified improvements and has provided recommendations to the processing late CRC as well as data collection/analysis.

Data associated with late 2024 CRCs received through May 15, 2025, were analyzed to determine the (1) amount of catch on late CRCs and (2) the timing of when late CRCs were received after the reporting deadline. Starting in 2024, late CRCs were batched (grouped with other CRCs daily or weekly depending on the time of year), counted, and stored by WDFW. Review of the 2024 late summer CRC data showed that 53% of late summer CRCs and 78% of the associated late catch (21,529 lbs.) were received by November 15th. The remaining 47% of late CRCs and 22% of associated late catch (6,072 lbs.) were received between November 15 and May 15 of the following year. Analysis of the 2024 late CRC data indicated that catch on late CRCs accounted for 2.8% of the total summer catch and 1.5% of the total winter catch. Further review of the 2024 late CRC data suggested that crabbers who turned in a CRC late in 2024 behaved differently from both respondents and non-respondents for that year; crabbers who turned in late CRCs reported higher average catches than non-respondents but lower average catches than on-time respondents. For 2024, these results indicate that catch associated with late CRCs falls into its own

category and will need to be accounted for or estimated independent of other CRC data. All 2024 catch data, including late CRCs, was not available in time for the 2025 management plan season so the 2024 harvest estimate used for management was derived from only on time; reports that were available prior to the entry of late card data.

The TWG produced recommendations to determine catch associated with late CRCs and to have this portion of catch contribute to the overall catch estimate. In the event that the complete late CRC data is not available for a given year due to limited staff resources at WDFW, an alternative approach to the estimation of late CRC catch is also provided.

Facts and Information Supporting TWG recommendations

- On average, each year WDFW receives approximately 5,970 summer late CRCs ($2.8 \pm 0.005\%$ of CRC sold annually; 2014 – 2024) and 555 winter late CRCs ($1.8 \pm 0.003\%$ of CRCs sold annually; 2014 – 2024 CRC sold).
- Late CRC data from summer of 2024 indicated there was a difference in catch per card between on-time and late-reported crabbers (3.7 crab per crabber for late vs 6.5 crab per crabber for on-time), but it is unknown how representative this result is for all years due to only one year of data being available. Recommendations for in-season estimates of late-card catch is currently based on this difference but more years of data are needed to validate this difference.
- A ratio-based preliminary catch estimate can reasonably be used to predict the number of cards that are expected to be turned in late for a year/season and to estimate catch per card for late CRCs received each season. This method is currently limited by only having one year of late catch data. Additional data is needed to assess performance of this model approach and determine if it is a valid method for estimating preliminary and/or final recreational catch associated with CRCs turned in late.

Recommendations

The TWG recommends providing a late online reporting option to increase the number of direct reports of late catch.

Currently, late respondent data only comes from CRC holders who physically mail in their CRCs after the reporting deadline.

TWG reiterates its recommendation from 2024 that catch from late CRCs be counted and entered into the CRC database; these data will be included separately from non-respondents in annual harvest estimates. Moreover, WDFW should continue to enter and count catch on CRCs received after the reporting deadline and include the complete late CRC catch in the final recreational catch estimates each year.

The TWG acknowledges that the most accurate catch estimate is one that accounts for all reported catch and concludes that catches contained on those late CRCs represent reported catch that should be incorporated into final harvest estimates by the end of the management year. Including the actual catch on late CRCs reduces the need to generate an estimate for a portion of the CRC holders.

Currently, only one year of late CRC *catch* data has been collected for each season (2024 summer and winter crab fisheries). This limited dataset does not provide enough information to observe consistent patterns in late CRC catch, either relative to other components of the catch (such as to on-time respondents or non-respondents) or in time (relative to other years of late CRC respondents). A minimum of five years of late CRC data are likely necessary to support such an analysis by the TWG. Continued entry of late CRCs will build a dataset that is needed to analyze trends of late respondents which could then be improve estimates of late card catch in the recreational catch accounting system.

TWG recommends using an estimation-based calculation methodology (see below) to provide in-season estimates *as a place holder* until the census of the late respondent catch data can be used in the final catch estimate.

A primary reason for late CRCs not being a part of the current harvest estimation framework stems from the delay in receiving and compiling data from late respondents to produce final harvest estimates. The TWG recommendations include methods to produce an ‘in-season’ estimate, which can be used as an in-season placeholder until the ‘final’ late respondent catch is generated using available late CRC data for that year and season.

TWG members agree that using a census or count of catch from late CRCs is the best way to account for recreational catch reported on late CRCs. Some members of the TWG supported an alternative (described below) to using the estimation-based calculation methodology for the final catch estimate, but the majority of TWG recommends waiting until at least five years of late respondent catch data is available to make a more robust estimate of mean catch per CRC from late respondents.

Methods are presented to generate a final catch that is a *census* of late CRCs (preferred option; counting catch from late CRCs as they come in throughout the season) or an alternative method that is an *estimate* of catch from late CRCs using the ratio-based method.

In-season Estimate Methods: *Ratio-based estimate of catch for late respondents*

A ratio-based preliminary catch estimate for late respondents is produced using the predicted number of late CRCs received for the season and the estimated catch per late respondent. This in-season estimate serves as a placeholder for in-season management.

‘In-season’ late CRC estimate would be calculated as:

$$\widehat{C}_L = \widehat{N} \times \widehat{\bar{C}}_L$$

Where,

$\widehat{C}_L$ = Estimated late CRC catch for a given year and season.

$\widehat{N}$ = Estimated number of late CRCs for the current year and season. Calculated as,

$$\widehat{N} = \# \text{ of CRC sold in a season} * R_{CRC}$$

Where R_{CRC} is calculated as the five-year running average of the ratio of late CRC turned in for the season to the total number of CRCs purchased for that season.

$$R_{CRC} = \frac{\text{Number of Late CRCs}}{\text{Total CRCs Sold}}$$

$\widehat{C}_L$ = Estimated catch per late respondent. Calculated as,

$$\widehat{C}_L = R_{Catch} * \text{Catch per ontime CRC}$$

Where R_{Catch} is the three-year running average of the ratio of catch per late CRC to the catch per CRC turned in on time.

$$R_{Catch} = \frac{\text{Catch per Late CRC}}{\text{Catch per ontime CRC}}$$

*Note - For 2025 and 2026, a three-year average for R_{Catch} will not be available, so the TWG recommends using the available data (one-year then two-year averages).

Final Estimate Methods (Preferred): Census catch

Complete late CRC data received through the cut-off date is summed and included in the final recreational estimate, which gets updated before the next management plan season.

$$C_L = \sum_{i=1}^{N_L} C_L$$

Where,

C_L = Catch per late respondent (for the current year)

N_L = Total late CRCs received in a season (for the current year)

For implementation, this method requires annual entry of CRCs received after the reporting deadline. Late CRC catch will be calculated for each year and season separately.

Final Estimate Methods (Alternative): Ratio-based estimate of catch for late respondents

**NOTE: this recommendation is not universally supported by all members of the TWG and will need more data to determine if it is an acceptable method.*

Late respondent catch estimate using the alternative method involves revising the ‘in-season’ estimate to include the complete number of late CRCs received that season through the cut-off date and producing a final recreational estimate prior to the next management plan season. For implementation, the alternative method does not require annual entry of CRCs received after the reporting deadline; however, the parameters of this model need to be updated with current data on a regular frequency (i.e., every other year). All late CRCs received through May 15 of the following year are scanned and retained by WDFW, so a ratio can be evaluated retrospectively.

The TWG recommends at least five years of late CRC data be entered to evaluate whether there is a predictable relationship between catch reported on CRCs received on-time versus those CRCs received late. Adoption of this method would need to be agreed to by co-managers before implementation.

Under this alternative method, the final late CRC catch estimate is calculated as:

$$\widehat{C}_L = N_L * \widehat{\overline{C}}_L$$

Where,

N_L = Actual number of late CRCs for the current year and season. $\widehat{\overline{C}}_L$, is the estimated catch per late respondent calculated as described above under the ‘in-season estimate section.

CATCH RECORD CARD NON-RESPONDENTS

Description

Estimation of non-respondent catch requires some level of information from CRC-holders who do not return their CRCs. Prior to 2010, phone surveys of non-respondents were conducted annually to acquire data from non-respondents, which was used to inform recreational harvest estimates. These types of surveys are costly and difficult to implement, so beginning in 2011, WDFW converted to using the NRBCM (commonly referred to as Bias Correction Model), to estimate non-respondent catch in lieu of conducting annual surveys.

The NRBCM is a quadratic regression model used to predict the bias correction needed based on the reporting rate (i.e., proportion of CRCs returned). The difference in average catch between respondents and non-respondents is known as “non-response bias”. At 100% reporting rate, no bias correction is required because all catch is reported on the CRCs but as the reporting rate decreases, the influence of the bias correction on the final catch estimate increases. Non-respondents are less likely to have participated in recreational harvest (approximately 24% retained crab in summer 2023) when compared to respondents (51% retained crab in summer 2023). NRBCM was originally derived from telephone survey data of non-respondents that occurred from 2007 – 2010. Recreational catch is corrected for non-response bias due to observed differences in behavior between those who return their CRC’s and those who fail to return their CRC’s (Tarrant et al. 1993). The portion of the population described as “non-respondents” has changed over time. Prior to 2025, all CRC holders that failed to submit their catch by the seasonal reporting deadlines were considered non-respondents. Following the work of the TWG, the definition of non-respondents changed to only include those CRC holders that failed to return their CRCs prior to May 15. CRC holders that returned their CRCs after the reporting deadline but before May 15 were designated as “late respondents” at that time.

Given the recommendations coming out of the TRP report (Hankin et al. 2021) and the co-manager concerns regarding the outdated data used for NRBCM, non-respondent surveys

were repeated for both the summer and winter in 2022 and 2023. Between 2020 and 2024, an average of 57% of CRC holders did not submit their CRCs to report their crab catch, overall (51% in winter, 58% in summer). In 2022, non-respondents comprised 58% of the total number of CRC holders for the summer fishery, and the catch associated with this group comprised 27% of the total recreational harvest (estimated from telephone and online surveys that occurred in 2022). After the TWG review of the CRC-based estimation process and the data from the recent year of surveys, the group has chosen to move away from the NRBCM and now recommends alternative methods for estimating catch for non-respondents.

Choosing a method for estimating the catch of the non-respondents has been a challenge because there are no catch data from this group against which estimation methods can be compared. With only a few years of non-respondent surveys conducted in the last 10 years, there is insufficient data to identify predictable changes over time or to explore correlations with the catch of respondents. Additionally, members of the TWG and Tribal policy have cited concerns over the quality of data provided by non-respondents during interviews. The methods proposed by the TWG balance concerns over non-response bias (e.g., non-respondents are less likely to have gone crabbing or did not catch Dungeness crab) and recall bias (decreased accuracy in catch reports when asked about crabbing trips months after the harvest occurred).

In concurrence with the Technical Review Panel (Hankin et al. 2021), the **TWG agrees that, under the current CRC-based catch estimation system, annual sampling of the non-respondent group for both summer and winter is the most technically sound way to collect data for generating catch estimates of this group. The TWG also recommends identifying and implementing approaches to increase the reporting rate to reduce the number of non-respondents for which catch must be estimated.** The TWG acknowledges that increasing the rate of catch reporting (reducing the number of non-respondents each year) is a primary solution to addressing the issues with Dungeness crab harvest estimation, as more than half of licensed card holders fail to report catch.

Facts and Information Supporting TWG Recommendations

- 2.8% and 4.5% of summer non-respondents were surveyed by telephone in 2022 and 2023. The proportion of surveyed non-respondents who read catch information directly from CRCs was 25.2% (232 successful crabbers) and 30.4% (336 successful crabbers) from summer non-respondent surveys conducted in 2022 and 2023, respectively (**Table 1**).
- The percent of crabbers who successfully catch and keep crab is less for surveyed non-respondents than for respondents (21% for non-respondents in summer 2022 vs 44% for respondents, 24% for non-respondents in summer 2023 vs 48% for respondents)
- Summer nonrespondent survey data from 2022 and 2023 showed that, on average, non-respondents who read directly from CRCs reported fewer successful trips and retained fewer crab overall than respondents in the same year; **Table 1, Figure 2**). The number of crabs kept per trip was statistically similar between the respondent and non-respondent reading catch directly from CRCs in 2022 and 2023, according to two-sided Welch's t-tests ($p < 0.0001$; **see Table 2 and Figure 2 below**). A Welch's t-test was used because the data have unequal variances, and because t-tests can be generally robust to departures from normality (as is the case for the trip data here). However, the Welch's test is not robust to all of the assumptions of parametric statistics, particularly when sample sizes are unequal; and so the TWG agrees that future review of the respondent and non-respondent data should include a determination of whether it is most appropriate to compare means or medians.
 - It is important to note that exploration of the data revealed a higher proportion of non-respondents reading directly from their CRCs catch more than 5 crab per trip, which may reflect crabbers reporting multiple trips as one (**Table 3**). However, given the current amount of non-respondent survey data available, it is impossible to determine if the small percentage of trips with higher catches represent actual daily harvest or harvest over multiple

trips. It is important to note that these findings are all based upon only two years of data (2022 & 2023) and focus on summer non-respondents. Analysis of summer respondent and non-respondent data was the focus for WDFW primarily due to the fact that ~88% (average summer catch proportion from 2020-2025) of the total season catch each year comes from the summer fishery. More data from the non-respondent population is needed to improve the accuracy of annual estimates of recreational harvest and future analyses should also include examination of winter data.

Table 1: 2022-23 Summer Nonrespondent Survey Summary – Summary of sample sizes and proportions for summer non-respondent surveys conducted in 2022 and 2023 that were used as a basis for developing the methods to estimate non-respondent catch.

Summer	2022	2023
Number of Nonrespondents	122,016	103,508
Number of interviews completed	3,467	4,684
% of Nonrespondents interviewed	2.84%	4.53%
Number of people interviewed who kept crab (successful crabbers)	921	1124
% surveyed crabbers who kept crab	26.6%	23.6%
Number of successful crabbers interviewed reading directly off cards	232	340
% Successful crabbers interviewed who read directly from CRC	25.5%	30.4%
% of all completed interviews who were successful and who read catch information directly from CRCs	6.7%	7.3%

Table 2: 2022-23 CRC-level summary – Comparison of the mean number of successful trips, mean catch per successful card, and mean Dungeness kept per successful trip between summer non-respondents (NR) and respondents (R) from 2022 and 2023. Data includes only non-respondents who read directly from CRCs. Data processing of non-respondent survey data is further detailed in Appendix 4. A two-sided Welch's t-test ¹ was conducted to evaluate the statistical difference in means between the respondent and non-respondent groups for each year separately. P-values less than $\alpha=0.050$ indicate a statistically significant difference between the groups. (CI; 95% confidence interval). Respondent and non-respondent data were non-normally distributed according to a QQ plot and have unequal variances (see Fig 2 and 3).

	Respondents ($\pm$ CI)	Non-Respondents Reading Off Cards	P-value
2022			
Number of successful crabbers		232	
Mean Trips per Successful Card	3.26(± 0.03)	1.92(± 0.20)	<0.0001
Mean Dungeness per Successful Card	12.77(± 0.14)	7.81(± 1.00)	<0.0001
Mean Dungeness per Successful Trip	3.91(± 0.01)	4.07(± 0.32)	0.329
2023			
	Respondents	Non-Respondents Reading Off Cards	P-value
Number of successful crabbers		340	
Mean Trips per Successful Card	3.13(± 0.03)	2.00(± 0.16)	<0.0001
Mean Dungeness per Successful Card	12.27(± 0.13)	8.05(± 0.82)	<0.0001
Mean Dungeness per Successful Trip	3.92(± 0.01)	4.03(± 0.14)	0.113

¹ The two-sided Welch's t-test was selected because the mean was the preferred parameter for the recommended methods for 2025. The TWG agrees that further review of the respondent and non-respondent data in future years to determine if the mean or median should be applied.

Table 3: 2022-23 trip-level summary – Summary of trip-level catch reports from respondents (summer 2021-2024) and non-respondents (from summer 2022 and 2023 surveys). CRC Year, license year; Group Label, Report year and group type (R-respondent, NR-non-respondent); Confirm Card, Yes indicates catch and trip data was lead directly off a CRC and no indicates the catch was reported from memory; <=5, number of trips reported with less than or equal to 5 crabs (legal limit); >5, number of trips reported where more than 5 crab were reported; >10, number of trips reported where more than 10 crab were reported; >10, number of trips reported where more than 10 crab were reported; %>5, percent of reported trips where more than 5 crab were reported; %>10, percent of reported trips where more than 10 crab were reported; %>15, percentage of trips where more than 15 crabs were reported.

CRC Year	Group Label	Confirm Card	<=5	>5	>10	>15	% >5	%>10	%>15
2020	R 2020	-	178958	99	3	0	0.0553	0.0017	0.0000
2021	R 2021	-	204975	85	8	3	0.0415	0.0039	0.0015
2022	NR 2022	no	1459	46	22	17	3.0565	1.4855	1.1518
2022	NR 2022	yes	431	8	1	5	1.8223	0.2315	1.1468
2022	R 2022	-	177641	48	0	0	0.0270	0.0000	0.0000
2023	NR 2023	no	1536	48	17	24	3.0303	1.0947	1.5385
2023	NR 2023	yes	665	8	4	2	1.1887	0.5979	0.2999
2023	R 2023	-	167003	39	1	7	0.0233	0.0006	0.0042
2024	R 2024	-	170015	44	0	0	0.0259	0.0000	0.0000

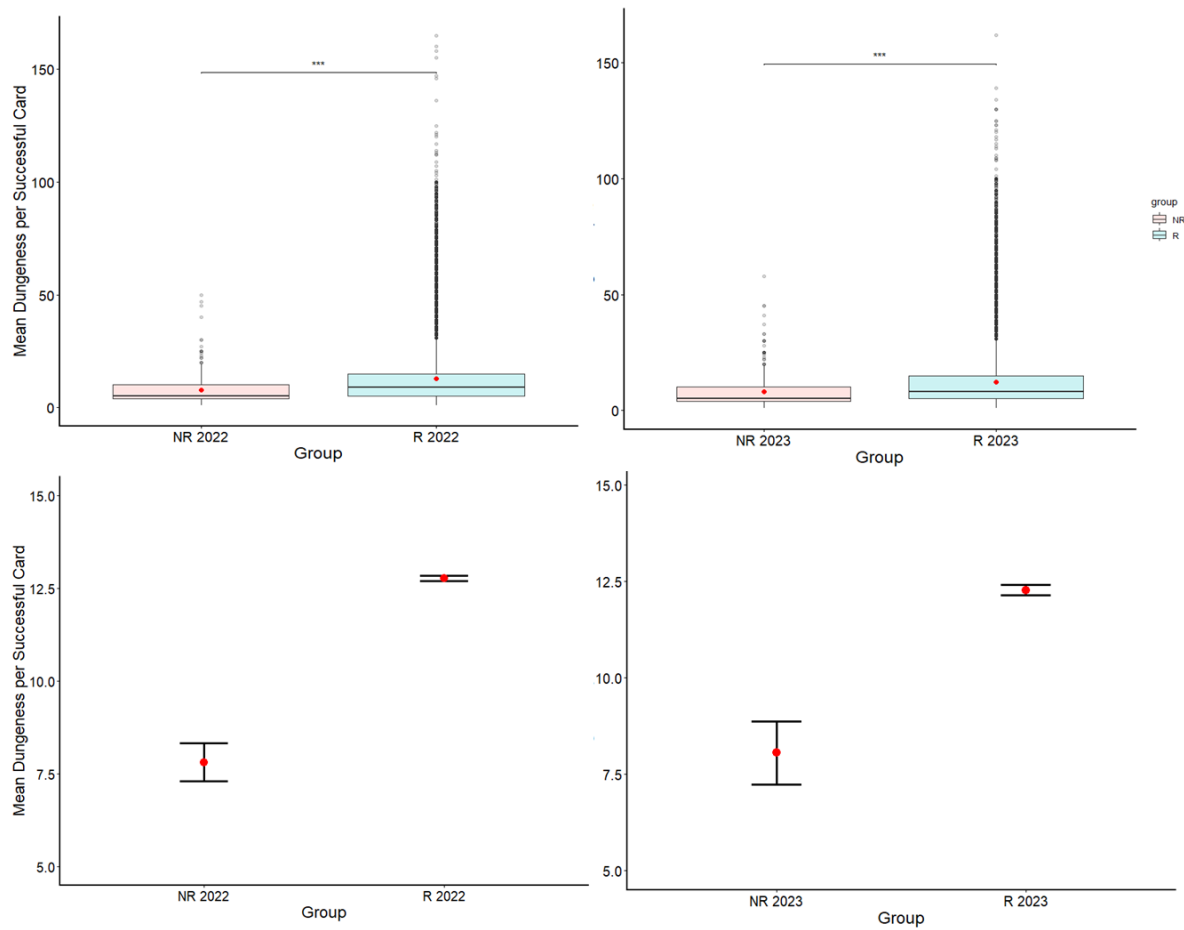


Figure 2: 2022-23 non-respondent to respondent mean catch comparison – Comparison of the *mean (red dot) Dungeness crab per successful card* between non-respondents (NR) and respondents (R) from 2022 and 2023. (Top) Boxplot showing the distribution of raw data from the non-respondent and respondent groups in each year. A Welch's t-test for differences in means indicated significant differences between the groups in each year (**Table 2**). (Bottom) The 95% confidence interval associated with the mean value for each group for each year. Data includes only non-respondents who read directly from CRCs.

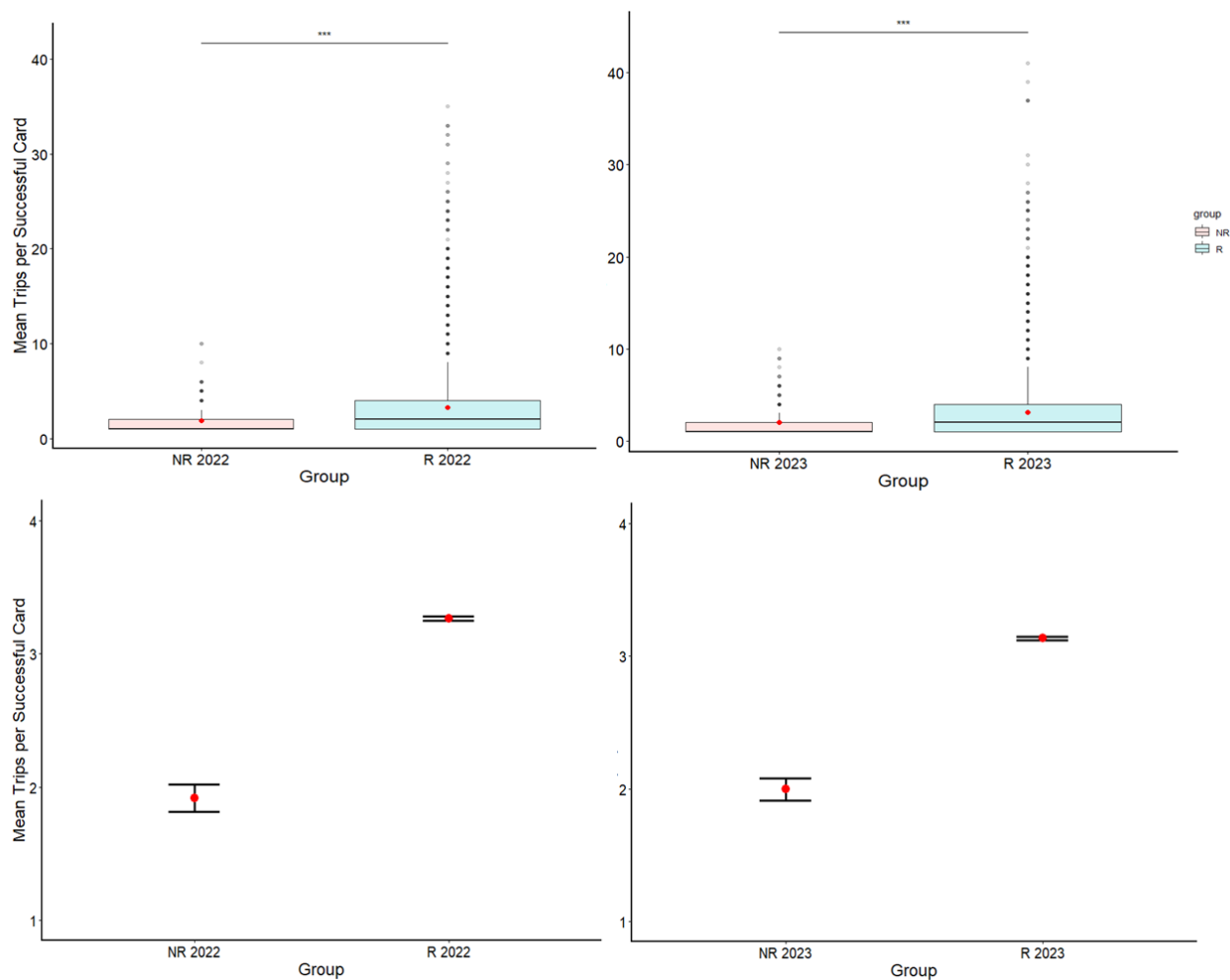


Figure 3: 2022-23 non-respondent to respondent mean trip comparison – Comparison of the *mean trips per successful card* (red dot) between non-respondents (NR) and respondents (R) from 2022 and 2023. (Top) Boxplot showing the distribution of raw data from the non-respondent and respondent groups in each year. A Welch's t-test for differences in means indicated significant differences between the groups in each year (**Table 2**). (Bottom) The 95% confidence interval associated with the mean value for each group for each year. Data includes only non-respondents who read directly from CRCs.

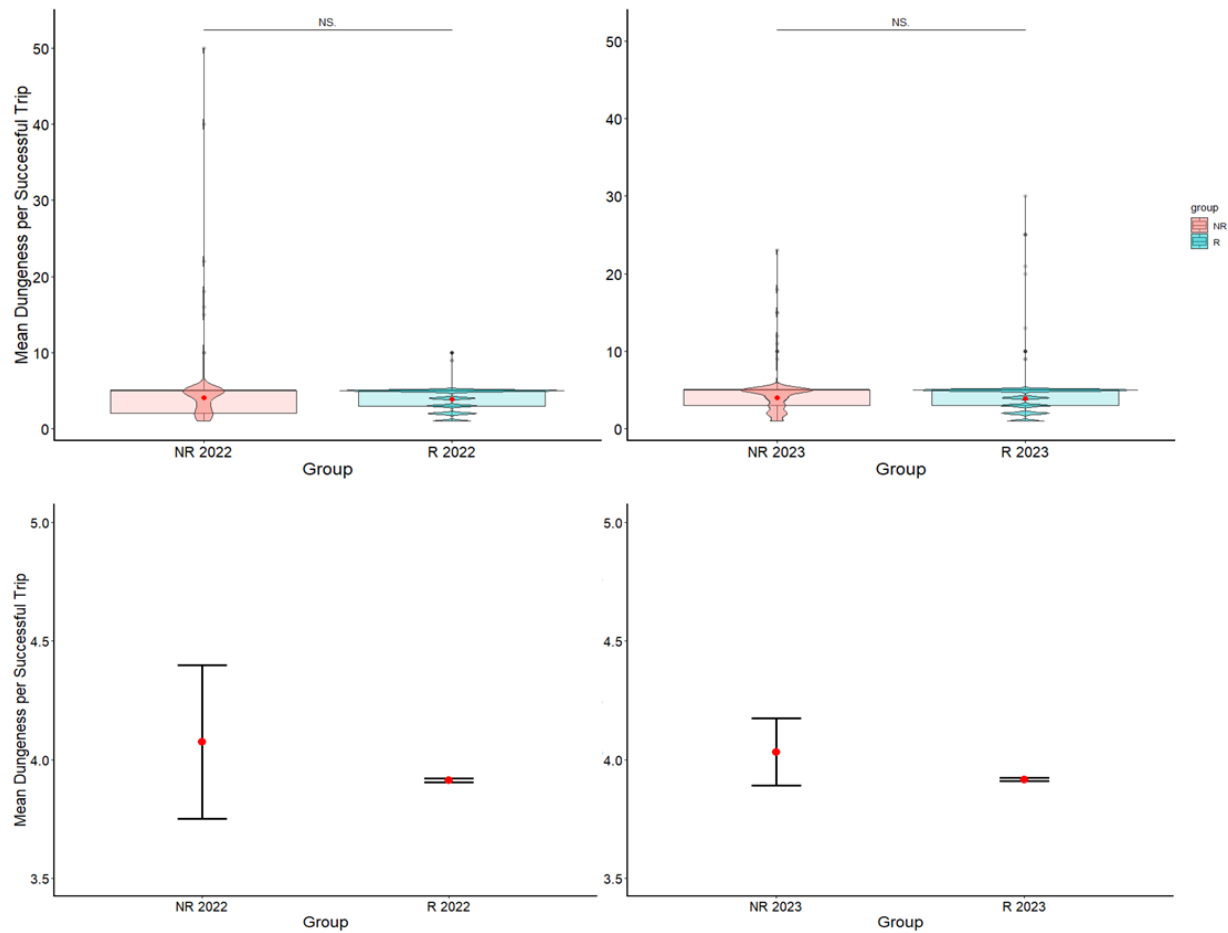


Figure 4: 2022-23 non-respondent to respondent mean crab per trip comparison – Comparison of the *mean Dungeness crab kept per trip per successful trip* between non-respondents (NR) and respondents (R) from 2022 and 2023. (Top) Boxplot, with overlaid violin, showing the distribution of raw data from the non-respondent and respondent groups in each year. A Welch's t-test for differences in means indicated significant differences between the groups in each year (**Table 3**). (Bottom) The 95% confidence interval associated with the mean value for each group for each year. Data includes only non-respondents who read directly from CRC from CRCs.

Recommendations

For 2025, since there are no plans to survey non-respondents, the **TWG recommended using a model-averaging between two specific models, the Catch per Card Method, and the Direct Report Catch per Trip Method**. Together, these approaches reduce recall and non-response bias in the non-respondent portion of the harvest estimate. For parameters that require non-respondent survey data, the **TWG recommends using the arithmetic mean of the values from the 2022 and 2023 non-respondent surveys for each corresponding year and season**.

Although this method was agreed-on by policy for the 2025 harvest estimates, upon deeper review of the data and code that was used as justification for the Direct Report Catch per Trip method, the TWG does not recommend continuing using this method due to the observation that a greater proportion of non-respondents reading from their cards catch more than 5 crabs per trip. Although it is not possible to confirm with the available survey data, a possible explanation is that non-respondents are condensing catch from many trips down to one trip to save time and this would affect the validity of the both the mean catch-per-trip and mean number of trips from that population. Despite uncertainty in the existing non-respondent catch data, some members of the TWG believe a trip-based approach, or similar method that captures potential differences in effort between the respondent and non-respondent groups, should be revisited in future iterations of NR surveys.

The TWG further recommends that future surveys (or other methods) used to collect catch data on non-respondents apply methods that ensure data align with assumptions of the estimation methods and can support model validation. **Information gained from future surveys can inform whether model averaging should continue or whether there is data to support selecting one model over another, or to apply a different method altogether based on available data and updated analyses.**

The TWG recommends conducting statistically robust surveys of non-respondents annually, and independently for each season and reducing recall bias by increasing the sample size of people reading the information directly from their unsubmitted CRC.

A description of the TWG recommended approach for estimating catch for non-respondents is provided below, with suggestions for future data collection.

Preferred: Model Average of the ‘Catch per Card’ and ‘Direct-Report Catch per Trip’ methods. This method will produce an estimate, and the arithmetic mean will be calculated to determine the overall catch of non-respondents.

The Catch per Card method

The *Catch per Card* method uses the mean percentage of crabbers that were successful from the 2022 and 2023 Responsive Management surveys and multiplies that by the number of non-respondents for the season (the total number of cards sold minus the number of respondents and the known or estimated number of late respondents). The number of successful non-respondents is then multiplied by the mean catch per card for successful on-time respondents for the season. Total catch for non-respondents using this method is calculated as,

$$\hat{C}_{NR} = \hat{p}_{NR} * N_{NR} * \bar{c}_R$$

Where,

$\hat{C}_{NR}$ = total estimated catch of non-respondents for a season

$\hat{p}_{NR}$ = estimated mean proportion of successful non-respondents who caught and kept crab; estimated from the most recent years of available non-respondent survey data (2- or 3-year average).

N_{NR} = total number of non-respondents for a season

$\bar{C}_R$ = mean catch per card from on-time respondents for the current year (mail and online reports).

Method Assumptions:

- Catch reported by respondents is truthful and accurate
- Non-respondents are truthful in stating whether they were successful while crabbing
- Average proportion of non-respondents that were successful (i.e., caught crab) in 2022, 2023 is representative of 2025.
- Catches reported by successful respondents are representative of catches of successful non-respondents
- Effort (i.e., number of trips) of successful respondents is representative of effort of successful non-respondents
- Average proportion of surveyed non-respondents that were successful is representative of the population of non-respondents that were successful.

Benefits:

- Doesn't rely on specific catch or effort data from the Responsive Management Surveys
- Simple and easy to understand and apply
- Limits the influence of recall bias by not relying on catch or effort data from the Responsive Management Surveys
- Uses the current year of catch rate data
- Potentially easier to implement future non-respondent surveys because less information needed from each non-respondent

The Direct-Report Catch per Trip method

The Direct-Report Catch per Trip method uses the mean number of trips by successful non-respondents *reading from their cards* by season from the 2022 and 2023 Responsive Management surveys and the mean catch per trip for the successful, on-time respondents for the season. Total catch for non-respondents using this method is calculated as,

$$\hat{C}_{NR} = \hat{p}_{NR} * N_{NR} * \hat{t}_{NR} * \bar{c}_R$$

Where,

$\hat{C}_{NR}$ = Total estimated catch of non-respondents for a season

$\hat{p}_{NR}$ = estimated mean proportion of successful non-respondents who caught and kept crab; estimated from the most recent years of available non-respondent survey data (2- or 3-year average).

N_{NR} = total number of non-respondents for a season

$\hat{t}_{NR}$ = estimated mean number of trips per non-respondent; estimated using the most recent years of non-respondent survey data from people who read directly from cards

$\bar{c}_R$ = mean catch per trip from on-time respondents from the current year (mail and online reports)

Method Assumptions:

- Catch and effort reported by respondents and surveyed non-respondents is truthful and accurate
- Non-respondents are truthful in stating whether they were successful while crabbing
- Catch proportion and trip data for non-respondents in 2022, 2023 are representative of 2025

- Catches per trip of successful respondents are representative of successful non-respondents
- Effort data (i.e., the number of trips) of surveyed non-respondents is representative of non-respondent population

Benefits:

- Doesn't rely on specific catch data from the Responsive Management Surveys
- Simple and easy to understand and apply
- Limits recall bias by excluding non-respondents without CRCs in front of them
- Uses the current year of per-trip catch rate data
- Accounts for differences in effort between respondents and surveyed non-respondents
- Potentially easier to implement future non-respondent surveys than asking for full catch and trip (date, location) details during the survey.

With annually collected data available, parameters of the above models could be replaced by values derived from the current year and season non-respondent survey data after positive technical review by WDFW and Tribal co-managers, and/or an independent reviewer.

If the non-respondent survey data is not collected annually, then the TWG recommends using the average of the most recent three years of available survey data (at a minimum) to estimate parameters used in the above models. Surveys of non-respondents should still be conducted regularly, and no more than two years should go by without a survey.

Considerations for Implementing Non-Respondent Surveys

Annual Surveys

Constraints: Requires significant staff and fiscal resources to conduct non-respondent surveys. Additionally, annual surveys will delay the time after each season before a final estimate can be produced. The methodology and sample size for conducting non-respondent surveys will need to be updated and agreed to.

Advantages: Would capture any annual variation in the proportion of catch attributable to non-respondents. Increasing the sample size of non-respondents reading directly from their card would reduce concerns about recall bias with the current data sets.

Surveys Conducted Less Frequently than Annually

Constraints: The proportion of non-respondents who caught and kept crab in the current year would be estimated based on previous years of available non-respondent data. There are not enough years of recent data on non-respondents to know how accurate a predictive model might be, so the TWG cannot currently quantify the additional uncertainty that using a multi-year average might introduce. The methodology and sample size for conducting non-respondent surveys will need to be updated and agreed to.

Advantages: The delay in producing final estimates would be reduced compared to the preferred option. The staff and fiscal cost associated with using a multi-year average rather than an annually conducted survey are significantly reduced.

Future Considerations

Further discussion is needed to refine future surveys of the non-respondent population, which should include evaluation of questions asked, survey design, and sample size. Surveys should be conducted using best practices. Survey needs for the two models

recommended by the TWG must determine: (1) whether the survey respondents read trip information from CRCs, (2) the proportion of non-respondents who are successful at crabbing in a given season and year, and (3) the mean number of trips taken by non-respondents in a given season and year.

The TWG acknowledges that these recommendations have been derived using the best information available to the whole TWG at the time of review. We recognize that as we continue to evaluate survey design and resulting data streams, we may continue to improve and refine recommendations for evaluating non-respondent catch.

These methods and data should be reviewed periodically and when data streams are added, subtracted, or altered (such as the roll-out of mobile electronic reporting, eCRC), to ensure best practices for establishing an estimate for non-respondent catch are current and continue to use the best available methods and data as agreed on by the TWG. Any future changes to these recommendations will require review and sign-off by members of the TWG. Additional topics that the TWG could cooperatively work on outside of the focus of recreational catch estimation have been compiled in a non-prioritized list within Appendix 6.

ESTIMATED UNREPORTED CATCH (EUC)

Description

Since 2016, all state and tribal crab harvests include a portion of catch that is not accounted for in the standard catch accounting process. This catch proportion, called the Estimated Unreported Catch (EUC), is added to the final harvest estimates. For recreational crab, the EUC accounts for crab that are harvested by CRC-holders but not recorded on a CRC, and for crab harvested by people that don't have a CRC (poaching). The current EUC value for recreational harvest is 9.97%, which was determined by an independent expert technical review panel (TRP) in 2021, who were tasked with deriving an appropriate method to estimate an EUC value with the data available at the time. The TRP used data from dockside creel interviews of recreational crabbers collected from 2010-2016 to derive the current Puget-Sound EUC value for the recreational fishery and established a method for revising the value when new data becomes available. The TWG reviewed the 2010-2016 data and found inconsistencies between the WDFW data and the data utilized by the TRP (details outlined in Appendix 4).

Following the TRP process and report (Hankin et al 2021), WDFW acquired resources to conduct crabber interviews on an annual basis and initiated a dockside creel program in 2023. This program has continued during each summer recreational crab season with limited opportunistic sampling occurring during the winter season in several Marine Areas. From its implementation in 2023, the sampling program consisted of eight 3-month sampling staff, who focused on collecting compliance information and biological data during the recreational crab fishery in Puget Sound. With limited staff available to conduct sampling, emphasis focused on intensive, random sampling of CRC holders using water access points that can be safely and logistically sampled in specific Marine Areas (MA) each year, with the goal of collecting data to inform Marine Area-specific EUC values. In 2023, the recreational crab creel survey did not follow a random survey design, instead that year's survey focused on access points where crabbing effort was generally observed

to be highest and was intended to guide a more statistically robust survey in future years. Furthermore, in 2023, interviews were conducted at the level of individual successful crabbers from each contacted vessel; however, in following years (2024 and 2025), interviews were conducted at the level of individual vessels, summing the total crab recorded across all CRCs holders on the vessel and the total crab retained per vessel.

Following 2023, the survey focused creel sampling effort within a few select Marine Areas in 2024 and 2025 (MAs 8-1 and 8-2 in 2024; MAs 8-1, 8-2, 9, and 11 in 2025) and followed a weighted random sampling survey design to attempt to sample more access sites within a Marine Area that were not previously collected in 2023, including private marinas and smaller public boat ramps. Although the weighted random survey design used in 2024 and 2025 is intended to sample CRC holders from both public and private water access sites, the survey does not sample those who access the water through sites that are logistically challenging to sample (e.g., shoreside homeowners).

In this document, focal-area sampling refers to data collected during creels in 2024 and 2025 in Marine Areas where a weighted random survey design was used, whereas data collected in 2023 and other Puget Sound Marine Areas in 2024 and 2025 which were sampled opportunistically are referred to as non-focal or 'Index'. Most open Marine Areas have been sampled to some degree each year since the study was first launched (Marine Areas 4 and 5 have not yet received dedicated sampling effort). After 3 years of sampling efforts, WDFW's recreational crab fishery sampling program has generated contemporary data needed to update the EUC values for the recreational crab fishery. The TWG has reviewed and discussed the available creel survey data from 2023, 2024, and 2025 and has provided recommendations for updating the EUC using methods that align with recommendations of the 2021 TRP report.

Facts and Information Supporting TWG recommendations

- Current probability-based focal area sampling data can be used to estimate EUC values in Marine Areas where sampling has occurred. Focus sampling was conducted in 2024 (8-1, 8-2) and 2025 (8-1, 8-2, 9, 11).
- For Marine Areas where focal area sampling has not yet occurred or is not currently planned (i.e., closed [Marine Area 13] or remote Marine Areas [Marine Areas 4 and 5]), EUC can be estimated by aggregating data collected from all Marine Areas.
- An agreed to analysis to determine the appropriate number of interviews to estimate Marine Area specific EUC values has not been conducted. It is unknown if focal or non-focal surveys have met an appropriate sample size target to generate an EUC for individual Marine Areas.

Recommendations

Over the last three years, the WDFW recreational crab sampling program has been developed to meet the long-term goal of producing data required to update annual Marine Area-specific EUC values. Given the current level of staffing and resources available for this project, the WDFW team chose to focus on sampling fewer Marine Areas intensively in a given year to produce information with the spatial and temporal resolution and sample size to support Marine Area-specific EUC estimates. Focal-area sampling has only occurred in four Marine Areas as of 2025 (MAs 8-1, 8-2, 9, and 11). However, the WDFW sampling plan for 2026 (and future years) will include Marine Areas 6, 7, 8-1, 8-2, 9, 10, 11, and 12 as a focus area following a simple or weighted random survey design. Recognizing that comanagers are interested in updating the EUC for the 2025 recreational harvest estimate, the TWG developed three alternative approaches that utilize the most recent available data. These approaches are also proposed as an interim solution for the 2025 recreational harvest estimate. Additionally, we provide recommendations for establishing a methodology to produce annual updates to the EUC using statistically robust methods.

Data Processing Recommendations

On January 8, 2026, the TWG met in Edmonds to discuss the creel data and decide on recommendations to process the creel data for estimates of EUC. The TWG discussed how to treat the following aspects of the creel data:

1. The TWG decided to retain all creel data for a Marine Area regardless of whether the interview occurred during a focal area survey or index survey.
2. The TWG discussed how to handle situations where no CRCs were checked during an interview, but other information was recorded, such as the number of crabs in possession. It was agreed that interviews where CRCs were not checked should be treated as incomplete and excluded from estimates of EUC. No assumptions are made about recording behavior for cases where the creel sampler could not check their CRCs.
3. Situations where the number of crabs recorded on a CRC is greater than the number of crabs in possession should be converted to match what was in possession.
4. The TWG decided to define a vessel contacted as the sampling unit and recommends using the number of successful vessels contacted with CRCs checked as the unit of weighting (when calculating the mean EUC across years).
5. The TWG recommended including winter season creel data for Marine Area 7, specifically any creel that occurred in Marine Area 7 during September but after Labor Day, which is the only area open during this time period and creels have occurred. As of 2025, Marine Area 7 has not had a focal survey and the period of September after Labor Day represents on average (2023-2025) 48% of the available creels for this area.
 - a. Following the January 8th meeting, WDFW reviewed the post-Labor Day MA7 September creel data and found inconsistencies in the data collection methods by the creelers during the years 2024 and 2025. As such, WDFW no longer supports including the post-Labor Day MA7 September creel data for

updating the EUC for 2025 recreational harvest estimate but supports collecting and including data collected during this time frame in future years following an update to the survey questions to avoid these inconsistencies. This position is not supported by all technical working group members.

2025 EUC Update

Due to the complexity and lack of parity in the creel data collected, the TWG developed a suite of options for calculating EUC from creel data. Adoption of recommendations will be made on a regional basis after co-manager agreement.

Option A: Puget Sound -Wide estimate for all Marine Areas using most recent 3 years of survey data

Apply an average Puget Sound-wide EUC estimate for all Marine Areas calculated using the most recent 3 years of sampling data for all Marine Areas (2023, 2024, 2025) A Puget Sound wide EUC value for each year is calculated as an average of the estimated EUC values for each Marine Area weighted by the proportion of reported harvest from CRCs by Marine Area. The overall Puget Sound-wide EUC estimate is calculated as an average of the three Puget Sound wide EUC values estimated for each individual creel year weighted by sample size (total number of successful crabbing vessels interviewed each year).

Justification for method: This option provides an alternative if the co-managers prefer a Puget Sound-wide estimate. Combining all data from all the years and Marine Areas (both focal and index methods) provides a Puget Sound-wide estimate that can be used as an interim value until focal sampling can be conducted in all Marine Areas.

Option B: Marine area-specific estimate for focal areas using current year focal data and Puget Sound wide estimate for non-focal Marine Areas

Focal Sampled Areas: MA 8-1, 8-2, 9, 11

Apply Marine Area-specific

EUC values to Marine Areas sampled with random probability-based methods using the most recent year of data collected (2025) and using the TRP recommended calculations (unrecorded/recorded).

Non-focal areas: MA 4, 5, 6, 7, 10, 12,13

Apply the methods used in Option A to calculate a Puget Sound wide EUC value for the non-focal areas.

Justification for method:

Focal Sampled Areas (MA 8-1, 8-2, 9, 11): It is reasonable to assume that the behavior of recreational fishers may differ from year to year and by Marine Area. With the sampling effort conducted for the focal-area survey design (e.g., 3,157 focal area interviews in 2024 and 2,787 focal area interviews in 2025), the 2025 data can be used estimate the Marine Area-specific EUC parameter for application to the current fishery year and provides a direct measurement of current fisher behavior (CRC recording compliance). Because this value is derived from sampling crabbers during the current fishery year, it does not need to be estimated using previous years of data, which may not reflect current behavior. The focal area sampling approach uses probability-based, stratified sampling and the *most recent* available data.

Non-focal Areas (MA 4, 5, 6, 7, 10, 12,13): Focal sampling has not yet been completed in all areas leaving several areas with a smaller sample size of creel interviews to generate an EUC estimate from. Combining all data from all the years and Marine Areas (both focal and index methods) provides an estimate of a Puget Sound-wide interim value that can be applied until focal sampling can be conducted in the remaining Marine Areas. Combining information from both focal and index creels will also supplement the information from Marine Areas that have not yet had a focal-area creel (e.g. someone who crabs in area 12 but who launches from a creeled site in area 9).

Option C: Marine Area-specific estimate for focal areas using recent 3-years of data and Puget Sound-wide estimate for non-focal areas

Focal Sampled Areas: MA 8-1, 8-2, 9, 11

Apply Marine Area-specific EUC values to focal areas using the most recent 3 years of data (combination of probability-based and opportunistic surveys 2023, 2024, 2025) and using the TRP recommended calculations (unrecorded/recorded; weighted average based on sample size [number of successful crabbing vessels contacted] by year).

Non-focal sampled areas: MA 4, 5, 6, 7, 10, 12, 13

Same as in Option B.

Justification for method:

Focal Sampled Areas (MA 8-1, 8-2, 9, 11):

The recent WDFW sampling program aimed to characterize crabber reporting behavior has only been active for 3 years, and over that time, multiple survey designs have been employed. As a result, there is limited information to inform whether observed differences between years and Marine Areas are due to true differences in crabber behavior or random noise in the data. Additionally, education on recreational harvest rules may occur following creel interviews, therefore it is possible that over the course of a sample year or between sampling years, the creel surveys themselves may influence the behavior of recreational participants at a given location without a commensurate change in behavior occurring at less sampled or un-sampled locations in a Marine Area, potentially influencing how representative a sampled creel site is of the broader crabber population.

Non-focal Areas (MA 4, 5, 6, 7, 10, 12, 13):

Same non-focal methods described in Option B.

Future Considerations

The future Puget Sound recreational crab catch accounting system will rely heavily on the creel sampling program to produce data and information to generate reliable and accurate harvest estimates each year. The recommendations above provide alternative options to address EUC in the short term, specifically for the 2025 recreational harvest estimate. The TWG has also identified additional recommendations, primarily focused on refining creel survey methods, to improve the current CRC-based harvest estimation system and to inform future harvest estimation methods. Additional topics that the TWG could cooperatively work on outside of the focus of recreational catch estimation have been compiled in a non-prioritized list within Appendix 6.

TWG general EUC/Creel survey recommendations

- 1. WDFW should consult with an expert on survey design to ensure creel methods follow suggestions by the TRP (2021) for unbiased sampling and that the methods produce reliable, accurate data needed to estimate management parameters at the appropriate scale.**
- 2. WDFW should consult with the tribes early in the survey design planning process to advise areas of focus and survey methods.**
- 3. WDFW should sample all Marine Areas each year with adequate intensity and appropriate methods to derive annual EUC estimates, as informed by a third-party expert and treaty co-managers.**
- 4. Future creel surveys and data processing decisions should consider applying EUC from portions of Marine Areas to Crab Management Regions where there is spatial mismatch between the two. For example, interviews of recreational crabbers that harvested in the 25C portion of Marine Area 9 should be applied to EUC calculations in Region 5 not Region 2W.**

5. **Review and consider sample size targets necessary for each Marine Area to update regional EUC values.**

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APPENDIX 1 – Joint Shellfish Technical Working Group Participants

Shellfish Technical Working Group Co-Chairs

Name	Affiliation
Katelyn Bosley	Washington Department of Fish and Wildlife
Galen Johnson	Northwest Indian Fisheries Commission

Shellfish Technical Working Group Section Leads

Name	Affiliation
Katelyn Bosley	Washington Department of Fish and Wildlife
Courtney Hart	Port Gamble S'klallam Tribe
Maddison Hicks	Swinomish Indian Tribal Community
Megan Russell	Lummi Nation
Elizabeth Tobin	Jamestown S'klallam Tribe
Daniel Sund	Washington Department of Fish and Wildlife
Don Velasquez	Washington Department of Fish and Wildlife
Blair Winnacott	Washington Department of Fish and Wildlife

Shellfish Technical Working Group General Members

Name	Affiliation
Sarah Anderson	Suquamish Tribe
Viviane Barry	Suquamish Tribe
Kendall Barton	Tulalip
Matt Beirne	Lower Elwha S'klallam Tribe
Marlene Bellman	Northwest Indian Fisheries Commission
Katelyn M Bosley	Washington Department of Fish and Wildlife
Jim Bura	Nooksack Indian Tribe
Aaron M Dufault	Washington Department of Fish and Wildlife
Mary Fisher	Swinomish Indian Tribal Community
Devynn Gately	Jamestown S'klallam Tribe
Alex Gouley	Skokomish Indian Tribe
Courtney Hart	Port Gamble S'klallam Tribe
Jason Haveman	Port Gamble S'klallam Tribe
Maddison Hicks	Swinomish Indian Tribal Community
Margaret Homerding	Nisqually Indian Tribe
Lane Jackson	Lower Elwha S'klallam Tribe
Galen Johnson	Northwest Indian Fisheries Commission
Aaron Jones	Tulalip
Eddie Kim	Squaxin Island
Michael Lennon	Washington Department of Fish and Wildlife

Nik Matsumoto	Suquamish Tribe
Mike McHugh	Tulalip
Oliver Miler	Northwest Indian Fisheries Commission
Shannon Miller	Point No-Point Treaty Council
Austin Paul	Point No-Point Treaty Council
Blair Paul	Skokomish Indian Tribe
Franchesca Perez	Stillaguamish
Annie Raymond	Jamestown S'klallam Tribe
Guy Redfox	Lummi Nation
Gordon Rose	Northwest Indian Fisheries Commission
Megan Russell	Lummi Nation
Heidi Rutherford	Washington Department of Fish and Wildlife
Clara Schlieman	Lummi Nation
Sarah Scott	Muckleshoot Indian Tribe
Eric Sparkman	Squaxin Island
Cathy Stanley	Tulalip
Ben Starkhouse	Lummi Nation
George Stearns	Puyallup Indian Tribe
Daniel M Sund	Washington Department of Fish and Wildlife
Elizabeth Tobin	Jamestown S'klallam Tribe
Don E Velasquez	Washington Department of Fish and Wildlife
David Winfrey	Puyallup Indian Tribe
Blair Winnacott	Washington Department of Fish and Wildlife
Sandy Zeiner	Northwest Indian Fisheries Commission

APPENDIX 2 – Fall 2024 Technical Working Group Recommendations

Joint Shellfish Technical Work Group Recommendations for Immediate Adoption September 17, 2024 - Final

Executive Summary for Policy:

Purpose/Approach

- This report summarizes a series of recommendations from the Joint Shellfish Technical Work Group (JSTWG) following a review of the current Catch Record Card (CRC) data entry and processing system currently in place to estimate the recreational harvest of Dungeness crab in Puget Sound.
- The listed JSTWG recommendations for the CRC data entry and processing system can be enacted immediately and applied in the generation of 2024-2025 recreational catch estimates.
- The JSTWG was formed to collaboratively resolve shellfish technical issues and develop recommendations to improve the management of shellfish resources in Puget Sound.
- The primary assigned task of the JSTWG is to build understanding, trust and identify areas of improvement for WDFW's CRC-based catch accounting system for the Puget Sound recreational Dungeness crab fishery. The group is currently working on this assigned task.

Findings

- WDFW's Puget Sound CRC data processing and storage system is decades old, uses outdated technology, and is housed in various disparate locations.
- A preliminary review of the current data entry and processing system revealed several small changes that can be enacted to improve the quality and integrity of the CRC data from mailed-in cards.
- In the last year, the WDFW crab management team has improved communication and collaboration with the WDFW CRC unit on the calculation of recreational estimates. This will improve transparency in the methods and allow for flexibility to adjust the calculations to ensure the most appropriate approach is used.
- In 2023, WDFW implemented an online survey approach to assess recreational catch in Marine Area 9 (25C) and the Region 3 subareas to inform the calculation of catch at the appropriate spatial scales for quota management. This approach can be applied to future seasons.
- Mailed-in CRCs are not retained and are shredded after catch records are entered. Cards that are received after the reporting deadline (late) are tracked but the data is not entered. Late cards are not retained and are also shredded.

Recommendations from the JSTWG for policy review and approval

CRCs

- Follow-up contacts for CRCs that are illegible or missing information should occur as soon as possible after they are received.
- Follow-up contacts for CRCs should follow communication guidelines to ensure non-leading questions are asked.
- Any follow-up contacts regarding CRC information should be tracked in a database (number of contacts, what information was added or altered), and any information not included on the originally submitted CRC should be noted (such as information filled in following contact with CRC holder or filled in due to assumption by CRC staff.)
- Conduct analyses to understand the trends and magnitude of late CRC reporting to inform whether or not the magnitude of late/shredded CRC's impacts catch estimate[†].
- The catch data on late CRCs should be recorded in the CRC database.
- The timing of batching and uploading late CRCs into the WILD system should be standardized so that the trends in late card submission can be evaluated.
- **CRCs shredding should be discontinued.** Determine a longer-term storage solution for all submitted CRCs.

Data aggregation

- Marine Area (MA) 8 catch reported on mail-in cards should be split into MA 8-1 and 8-2 according to the percentages reported during that season.
- Online crabber survey data should be used to separate MA 9 into catch within CA 25C and catch from MA 9 outside CA 25C.
- Online crabber survey data should be used to apportion catch into subregions 3-1, 3-2, and 3-3.
- Starting in 2024, the online crabber survey should be extended to also apportion catch from MA7 into subregion 3-1.
- Catch should be aggregated into open season, closed season, and unknown for each catch area and season. Discontinue aggregation of catch into monthly catches. Discontinue moving catch from a closed season to an open season (as is done for Marine Area 7 winter harvest).

Summary of technical work to-date

The technical work group is comprised of WDFW, tribal and NWIFC technical staff and is co-chaired by Galen Johnson (NWIFC) and Katelyn Bosley (WDFW). The group has met four times since it was initiated in spring 2024 (5/23, 6/25, 7/31, 8/15). The goal of the initial round of meetings is to build an understanding of the underlying data and processes used to generate the recreational estimates and to identify recommendations for improvement that are supported by the entire

comanager group. To date the JSTWG discussions have built understanding surrounding aspects of the CRC system that were previously unknown or poorly understood.

Progress has included:

- Development of structure and function of the JSTWG to ensure engagement and collaboration from all members and distribution of workload.
- WDFW providing a deep dive into CRC recording, reporting, data entry, QA/QC procedures.
- Consensus surrounding several technical recommendations to conduct additional analysis on CRC data, enact modifications to the current CRC data workflow, improve data storage and documentation, adjust summarization methods, and develop new data sources to support catch estimation. Multiple documents have been finalized which contain these recommendations.
- Several small workgroups have been formed to begin making progress on priority tasks identified by the JSTWG. These workgroups will be focused on code translation, database development, and methods exploration for addressing non-response bias. These small workgroups will kick-off work during a series of meetings scheduled for 9/25.

Definitions

[†] Late CRCs are defined as CRC that are received by WDFW after the official reporting deadline has closed. Late-reported CRCs are primarily paper cards that are mailed to the department.

APPENDIX 3 – Fall 2025 Technical Working Group Policy Briefing

Technical Working Group Recommendations for Catch Record Card-Based Puget Sound Recreational Catch Estimation

11/5/2025

BACKGROUND

Following a series of meetings in 2024 and 2025, the Joint Shellfish Technical Workgroup (TWG) is providing initial recommendations to modify and update high priority components (see Figure 1) of the Puget Sound recreational Catch Record Card (CRC)-based methods by focusing on the following discrete topics:

- **RESPONDENT REPORTING AND GENERAL RECOMMENDATIONS** – Evaluating and updating data entry restrictions to online reporting, verification of missing/illegible information and clear documentation of CRC processing procedures.
- **LATE CATCH RECORD CARDS** – Evaluating and updating the recording and incorporation of harvest data submitted to WDFW after the CRC submission deadline directly into the harvest estimate.
- **CATCH RECORD CARD NON-RESPONDENTS** – Validating, documenting, and updating the methods used to estimate catch and effort by those recreational CRC holders that obtain a CRC but do not return them to WDFW.
- **ESTIMATED UNREPORTED CATCH (EUC)** – Evaluating the data available to inform the portion of catch harvested but not reported in the recreational crab fishery and establishing recommended procedures for regularly updating the EUC proportion using the methods described in the court-adjudicated Technical Review Panel (TRP) report (Hankin *et al.* 2021).

The purpose of this report is to detail recommendations made by the TWG specific to each topic for review by WDFW and Tribal co-manager policymakers. Terms are defined in a glossary at the end of the document.

The current CRC-based framework has been used to estimate recreational Dungeness crab catch in Puget Sound since 2007. Participants in the Puget Sound recreational Dungeness crab fishery are required by Washington State law (WAC [220-310-020](#) and WAC [220-310-010](#)) to submit CRCs detailing their catch by date and Marine Area and, if no catch occurred, submit a CRC with no catch. From 2020-24, an average of 43% of CRC holders returned their cards by the reporting deadline (49% in winter and 42% in summer.)

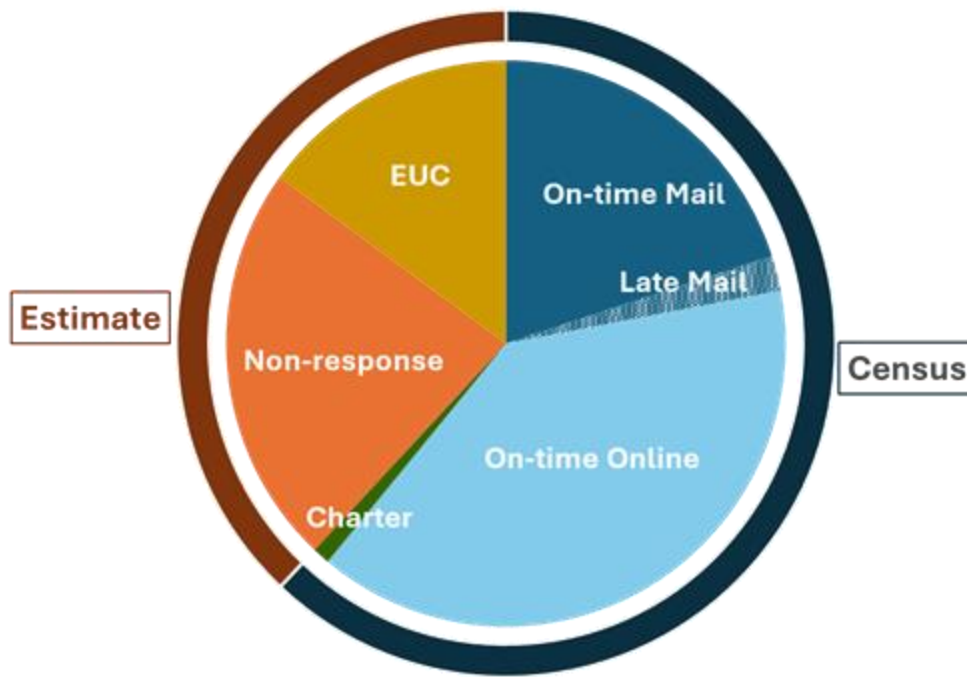


Figure 1: A pie chart representing each component contributing to the Puget Sound recreational Dungeness crab catch estimate. Pie wedges are not drawn to scale, as some portions are unknown and vary from year to year. The components of the harvest estimate that can be counted directly from CRCs are identified as census, and the components that cannot be counted directly (EUC and non-respondent catch) must be estimated.

Here, for policy consideration, the TWG offers its recommendations; there may also be minority reports provided by parties who do not agree with any of the recommendations. The Puyallup Tribe technical staff participated in the workgroup but does not agree with any of these recommendations. If a specific recommendation does not have consensus agreement by the entire TWG minus Puyallup, that is also noted under that recommendation.

ON-TIME RESPONDENTS AND GENERAL RECOMMENDATIONS

A large component of the catch estimate comes from recreational crabbers who purchased a CRC and submitted their CRC catch information to WDFW on or before the reporting deadline. The TWG completed a review of the data entry and processing procedures of CRC on-time reports in 2024 and 2025. The group presented some of their recommendations to policy in September of 2024 and many of these have been implemented. However, there are several recommendations for

on-time respondents and the overall recreational crab estimate that need to be implemented or completed. **The TWG recommends that WDFW:**

- **Allow more than 5 crabs to be reported per trip on the online reporting platform.**
- **Clear notation of which entries in the CRC database are based on crabber's recall through follow-up by CRC staff (e.g. for illegible and missing data on mail-in CRCs).**
- **Document current and past Dungeness crab recreational catch estimation methods.**
- **Consider approaches to increase the reporting rate to reduce the number of non-respondents for which catch must be estimated.**
- **Engage in periodic review of data collection and estimation methods with Tribal Co-managers to ensure that methodologies continue to be technically sound.**

LATE CATCH RECORD CARDS

Each year, some state recreational crabbers mail their CRCs to WDFW after the reporting deadline (an average of ~3% of summer CRCs and ~2% of Winter CRCs; 2014-2024). Prior to 2024, WDFW destroyed the late CRCs without recording their catch data. Catch from late CRCs returned by mail in 2024 was entered by the WDFW's CRC team, but not used for the 2024 recreational catch estimate.

The TWG has produced recommendations to determine catch associated with late mail-in CRCs and to have this portion of catch contribute to the overall catch estimate. If the complete late CRC catch data is not available for a given year an alternative approach to the estimation of late CRC catch is also provided. That alternative is supported by only a few parties to the TWG for use before three years of late catch are available. Most of the TWG is willing to support the method if 5 or more years of late CRC catch data is available to test the method.

The TWG recommends providing a late online reporting option to increase the number of direct reports of catch.

The TWG reiterates its recommendation from 2024 that late CRC catch be entered into the CRC database and that a census of that late CRC catch be included in the final recreational harvest estimate.

Complete late CRC data received through the cut-off date of May 15 the following year should be summed and included in the final recreational estimate.

$$C_L = \sum_{i=1}^{N_L} c_L$$

Where,

C_L = Total late CRC catch for a given season

c_L = Catch per late respondent i

N_L = Total number of late CRCs received in a given season

WDFW co-managers provide an in-season estimate to the Tribes in November to track harvest and help shape the fisheries for the rest of the year. Because late CRC catch is not yet available at this time, a method to predict estimated late card catch, as a placeholder, is needed.

The TWG recommends the following methodology for an in-season estimate until the total late respondent catch can be used in the final catch estimate.

A ratio-based preliminary catch estimate for late respondents should be produced using the predicted number of late CRCs received for the season multiplied by the estimated catch per late respondent CRC. This in-season estimate serves as a placeholder for in-season management.

In-season late CRC estimate would be calculated as:

$$\hat{C}_L = \hat{N}_L * \hat{c}_L$$

Where,

$\hat{C}_L$ = In-season estimated late CRC catch for a given season

$\hat{N}_L$ = Estimated number of late CRCs for a given season, calculated as the number of CRCs sold * a five-year running average of the ratio of late CRCs for the season to the number of CRCs purchased for the season.

$\hat{c}_L$ = Estimated mean catch per CRC from late respondents, calculated as a running three-year average of (catch per late CRC/catch per on-time CRC) * catch per on-time CRC for the current year.

Note - For 2025 and 2026, a three-year average for catch per CRC ratio will not be available, so the TWG recommends using the available data (one-year and/or two-year averages) only for those two years.

Some members of the TWG recommend an alternative to using the census catch for the final catch estimate from late CRCs for immediate use, but the majority of the TWG recommends

waiting until at least three years of late respondent catch data are available to make a more robust estimate of mean catch per CRC from late respondents.

In the alternate method, the in-season late respondent catch estimate is revised using the complete number of late CRCs received that season through the cut-off date for the final recreational catch estimate. The TWG supports entering late catch data at least every two years to keep this method current.

Under this alternative method, the final late CRC catch estimate is calculated as:

$$\hat{C}_L = N_L * \hat{\bar{c}}_L$$

Where,

$\hat{C}_L$ = Estimated late CRC catch for a given season

N_L = actual number of late CRCs for the current year and season.

$\hat{\bar{c}}_L$ = Estimated mean catch per CRC from late respondents, calculated as a three-year average of (catch per late CRC/catch per on-time CRC) * catch per on-time CRC for the current year.

Note - For 2025 and 2026, a three-year average for catch per CRC ratio will not be available, so the proponents of using the method during those years recommend using the available data (one-year and/or two-year averages). The proponents of waiting for three or more years of data has been collected to use this method do not support the one or two-year averages.

CATCH RECORD CARD NON-RESPONDENTS

In concurrence with the Technical Review Panel (Hankin et al. 2021), the **TWG agrees that, under the current CRC-based catch estimation system, annual sampling of the non-respondent group for both summer and winter is the most technically sound way to collect data for generating catch estimates of this group.**

For 2025 and beyond the TWG recommends a model-averaging between two models that help account for non-response bias and recall bias, (1) the *Catch per CRC Method* and (2) the *Direct Report Catch per Trip Method*. For parameters that depend on non-respondent survey data, the TWG recommends using the arithmetic mean of the two values for each year and

season from the non-respondent survey data from the 2022 and 2023 Responsive Management surveys for 2025.

Information gained from future surveys can inform whether model averaging should continue or whether there is data to support selecting one model over another.

A non-respondent catch estimate will be produced by taking the arithmetic mean of the following two methods for each season and using that in the in-season and final catch estimates.

(2) The Catch per CRC method

The Catch per CRC method uses the mean proportion of crabbers that were successful (caught and retained Dungeness crab) from the 2022 and 2023 Responsive Management surveys or future surveys and multiplies that by the number of non-respondents for the season (the total number of CRCs sold minus the number of respondents and the known or estimated number of late respondents). The number of successful non-respondents is then multiplied by the mean catch per CRC for successful on-time respondents for the season. Total catch for non-respondents using this method is calculated as

$$\hat{C}_{NR} = \hat{p}_{NR} * N_{NR} * \bar{c}_R$$

Where,

$\hat{C}_{NR}$ = Estimated catch of non-respondents for a season

$\hat{p}_{NR}$ = estimated proportion of successful non-respondents (those who caught and kept Dungeness crab), estimated from the most recent years of available non-respondent survey data (2-year average for 2025, 3-year average once new survey data is available)

N_{NR} = total number of non-respondents for a season

$\bar{c}_R$ = mean catch per CRC from all on-time respondents for the current season

(2) The Direct-Report Catch per Trip method

The 'Direct-Report Catch per Trip' method uses the mean number of trips by successful non-respondents *reading from their CRCs during a non-respondent survey* by season from the 2022 and 2023 Responsive Management surveys or future surveys and the mean catch per trip for the successful, on-time respondents for the season. Total catch for non-respondents using this method is calculated as:

$$\hat{C}_{NR} = \hat{p}_{NR} * N_{NR} * \hat{t}_{NR} * \bar{c}_R$$

Where,

$\hat{C}_{NR}$ = Estimated catch of non-respondents for a season

$\hat{p}_{NR}$ = estimated proportion of successful non-respondents (those who caught and kept Dungeness crab), estimated from the most recent years of available non-respondent survey data (2-year average for 2025, 3-year average once new survey data is available)

N_{NR} = total number of non-respondents for a season

$\hat{t}_{NR}$ = estimated mean number of trips per non-respondent; estimated using the most recent years of non-respondent survey data from people who read directly from their CRCs (2-year average for 2025, 3-year average once new survey data is available)

$\bar{c}_R$ = mean catch per trip from all on-time respondents from the current season.

The TWG recommends conducting statistically robust surveys of non-respondents annually for each season and reducing recall bias by increasing the sample size of people reading the information directly from their unsubmitted CRC. In the above methods, $\hat{p}_{NR}$ and $\hat{t}_{NR}$ averages could be replaced by an estimate derived from the non-respondent survey result for the current season after positive technical review by WDFW and Tribal co-managers, and/or an independent reviewer.

If the non-respondent survey data is not collected annually, then the TWG recommends using the average of the most recent three years of survey data to estimate parameters used in the above models. Surveys of non-respondents should still be conducted regularly, and no more than two years should go by without a survey.

Non-respondent Survey Considerations

Further discussion is needed to refine future surveys of the non-respondent population which should include evaluation of questions asked, survey design, and sample size. Surveys should be conducted using best practices must determine the following information to support the recommended models: (1) whether the survey respondents read trip information from CRCs, (2) the proportion of non-respondents who are successful at crabbing in a given season and year, and (3) the mean number of trips taken by non-respondents in a given season and year.

Estimated Unreported Catch (EUC)

The TWG agreed that EUC should be updated for 2025 and beyond but did not reach a consensus recommendation for that update. The TWG offers three options, which have varying degrees of support, in randomly selected order.

Option A: Apply a Puget Sound-wide EUC estimate for all areas using most recent 3 years of creel data.

Apply an average Puget Sound-wide EUC estimate for all areas, calculated using the most recent 3 years of creel data (for 2025, this would be 2023, 2024 and 2025), with each year weighted by the number of interviews as done for $\hat{R}_3$ from Hankin *et al* (2021). Because some areas have received much more intensive sampling than others, the TWG recommends a slight variation of this method in which the EUC is also weighted by area, with weighting based on the CRC catch proportion by Marine Area from those same three years. Weighting by area needs to be tested before this method is implemented; the TWG suggested some ideas if it does not work.

Option B: Area-specific estimate for focal areas using current year focal data and Puget Sound wide estimate for non-focal areas.

Focal Sampled Areas: MA 8-1, 8-2, 9, 11

Apply area-specific EUC values to areas sampled with random probability-based methods using the most recent year of data collected (2025) and using the TRP recommended calculations (unrecorded/recorded).

Non-focal areas: MA 4, 5, 6, 7, 10, 12, 13

Use the methods described in Option A.

Option C: For areas that have had a focus creel in the past three years, apply an area-specific estimate that uses the most recent three years of creel data for the area. For areas that have not had a focus creel in the past three years, apply a Puget Sound-wide estimate (as explained in Option A).

Areas that have had a focus creel in the last three years (MA 8-1, 8-2, 9, and 11 for 2025):

Calculate an EUC using the most recent 3 years (for 2025, this would include creel data from 2023, 2024 and 2025) of creel surveys of crabbers who reported catch in the area using the TRP recommended calculations ($\hat{R}_3$ from Hankin *et al* 2021). This would include information from crabbers who were part of the focus creels and those who were part of the other creels but who caught crab in the focus creel area.

Areas that have not had a focus creel in the last three years (MA 4, 5, 6, 7, 10, 12, 13 for 2025):

Use the methods described in Option A.

Finally, the TWG recommends that WDFW incorporate the following for general improvements of the creel surveys used to inform the EUC calculations:

- **Consult with an expert on sampling design to ensure that creel methods support the production of reliable, accurate, and unbiased EUC estimates.**

- **Consult with the tribes early in the creel survey planning process on areas of focus and survey methods.**
- **Creel surveys should minimize influence on the parameters being measured; education about recording catch should not be carried out during creel surveys that gather information on recorded and unrecorded catch.**

REFERENCES

Hankin, D., Armstrong, D., & Sampson, D. (2021). *Report of the Technical Review Panel: Unreported Catch in the Puget Sound Recreational Fishery for Dungeness Crab.*

GLOSSARY

Catch: Dungeness crabs retained by recreational crabbers, with or without a catch record card (CRC)

CRC: Catch Record Card. A mandatory catch reporting tool for recreational crabbers in Puget Sound. CRCs are issued for summer and winter seasons and must be submitted by a set deadline, either online or by mail, to avoid penalties.

Respondent: a CRC holder who returns their card and whose catch is directly censused.

On-time Respondent: A CRC holder who turns in their CRC data to WDFW online via WILD or email, by phone, by mail, or by drop-off before the official reporting deadline (October 1 for Summer, February 1 for Winter)

Late Respondent: A CRC holder who turns in their CRC data to WDFW via mail between the official reporting deadline and the late reporting cut-off date the following year (May 15th).

Non-respondent: A CRC holder who does not turn in the CRC data to WDFW by May 15 the following year.

Late CRC: A CRC turned in after the seasonal reporting deadline but before the late reporting cut-off date of May 15 the following year.

EUC: Estimated Unreported Catch (also sometimes referred to as Estimated Unrecorded Catch). The catch that is retained but not reported on a CRC, both by CRC holders and by non-CRC holders.

Recorded Catch: Catch that is marked on a CRC, CRC's returned without catch marked or reported online with zero harvest are recorded as zero catch.

Unrecorded Catch: Catch that is not marked on a CRC.

Creel survey: An in-person survey in which an interviewer asks a harvester about their fishing experience and collects information about the catch.

Focus area Creel: Focus area creels are designed to capture and describe the fishing behavior of crabbers within a designated Marine Area by intensive sampling at both high and low use water access locations that are currently practical to sample. On each open fishery day creel locations and a corresponding time of day are randomly selected based on a weighted method.

Index/Opportunistic Creel: An index creel is not scheduled; rather, it is sampled on an opportunistic basis contingent on the availability of additional creel staff. Index creels are designed to gather data and maintain a presence in Marine Areas that are not being sampled intensively or to explore water access points that have not been assessed by the creel program.

Marine Area: Marine Areas are waters within the boundaries of Washington, including Puget Sound, Hood Canal, the Strait of Juan de Fuca, the San Juan Islands, the Strait of Georgia, and the Pacific Ocean, and are the level at which recreational crab rules and regulations are set for the estimates made in this document. Marine Area boundaries are described in [WAC 220-311-010](#).

Crab Management Region: Waters within the boundaries of Washington, including Puget Sound, Hood Canal, the Strait of Juan de Fuca, the San Juan Islands, which directly relate to areas of individual management plan agreement with Treaty co-managers. Crab Management Regions are also defined in [WAC 220-320-110](#).

CRC database (also called catch database): a central repository for collocating catch reporting data from the various reporting sources which allows WDFW to process, perform analysis and generate estimates from a standardized data set

TWG: Joint Shellfish Technical Workgroup, consisting of shellfish biologists, harvest managers and biometricians from the Washington Department of Fish and Wildlife (WDFW), participating Treaty Tribes, and Northwest Indian Fisheries Commission (NWIFC)

WDFW: Washington Department of Fish and Wildlife

NWIFC: Northwest Indian Fisheries Commission

Nonresponse bias: The bias that occurs when respondents do not accurately represent the full population being sampled; in fisheries, nonrespondents tend to fish less than respondents.

Recall bias: The bias that occurs due to inaccurate recollection of past events; recall bias tends to increase the longer the time between the event and the recollection.

APPENDIX 4 – Updated Nonrespondent Survey Data Processing Rules

Responsive Management 2022/2023 Non-respondent Survey Data Processing Considerations – 05/19/2026 Update

Lead Authors: Lucas Rabins (Tulalip), Blair Winnacott (WDFW), Katelyn Bosley (WDFW)

1: Interviews recorded as “caught and retained crab” but with 0 crab recorded:

Seven interviewed non-respondents in the summer 2022 data file that harvested 0 crab were labelled as having ‘caught and retained’ crab. These individuals should be removed from the ‘trips’ dataset and converted from a ‘yes, I caught and retained crab’ to a ‘no, I did not catch and retain crab’ in the full data frame under ‘var619’. This is important if calculating the proportion of non-respondents that caught and retained crab. This would reduce the number of harvesters from 933 to 926.

Recommendation: Change these individuals to ‘no, did not catch and keep crab’ in the full data frame and remove them from the trip’s dataset.

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt this recommendation for processing the 2022 and 2023 Responsive Management telephone/email survey data.

2: Crab coded with “Other” location:

In the 2022 survey, non-respondents who did not recall the area of harvest but were able to describe the area with landmarks, the area was input as ‘Other’ (code 12) with an additional text column describing landmarks where the fishing took place. Many of these areas described correspond to true Puget Sound marine areas while some are coastal marine areas. Currently, these records are kept as “other”, but it is possible to look at the described area and manually enter the appropriate marine area and eliminate the records caught on the coast.

Example 1: Non-respondent Vcid ‘114375468’ input ‘other’ as the area of harvest then described their area by stating ‘Sequim Bay’.

Example 2: Non-respondent Vcid ‘114375468’ input ‘other’ as the area of harvest then described their area by stating ‘Westport’.

In the 2023 survey, additional options were provided to allow survey respondents to better partition out areas of harvest from areas inside and outside of Puget Sound. Options included: all marine areas (codes 1-11), coastal marine areas (code 12), area outside Puget Sound (code 13), ‘Other’(code 14) and ‘Don’t know’ (code 15). Unlike the 2022 data, data from the 2023 survey **did not** include the text column where respondents could provide a written description with landmarks indicating the area harvest.

Recommendation for 2022 data: Retain all ‘Don’t Know’ entries as unknown area. In both the full data frame and trips data frame, create a new column and manually enter the corrected marine

area based on the written area description for each trip listed with area code 12 (Other area). Interviewed non-respondents that harvested **only** in coastal areas should be converted from a 'yes, I caught and retained crab' to a 'no, I did not catch and retain crab' in the full data frame under 'var619'. Non-respondents that possess a Puget Sound Catch Record Card but harvested only in coastal areas should be treated as a 0-catch card. This would further reduce the number of harvesters from 926 (after implementing recommendation 1) to 921. Any trips that occurred in an area outside Puget Sound should be removed from the trips dataset.

Recommendation for 2023 data: Retain all 'Other' and "Don't Know" entries as unknown areas. Interviewed non-respondents that harvested **only** in coastal areas (code 12) or areas outside Puget Sound (code 13) should be converted from a 'yes, I caught and retained crab' to a 'no, I did not catch and retain crab' in the full data frame under 'var619'. Non-respondents that possess a Puget Sound Catch Record Card but harvested only in coastal areas (Code 12) or outside Puget Sound (code 13) should be treated as a 0-catch card. Any trips that occurred in an area outside Puget Sound should also be removed from the trips dataset.

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt these recommendations for processing the 2022 and 2023 Responsive Management telephone/email survey data.

3: Non-whole numbers for crab caught and date fields:

In the 2022 data, a single non-respondent recorded 3 trips with the amount of crab retained as 1.99. For these same trips, the day field was entered in as 1.31. Additionally, for 2 other trips from separate fishers, the day field is 16.23 and 18.192. All of these responses were entered in via the internet. There are several other similar non-whole number day fields in the 2023 data. In the document where RM responded to questions about this dataset, they said that the value of 1.99 crab retained was close to 2 and should be treated as 2 (rounded to the nearest whole number).

Recommendation: Convert decimal value entries for crab retained to 99 (unknown # crab) and day fields to 32 (unknown day).

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt these recommendations for processing the 2022 and 2023 Responsive Management telephone/email survey data.

When responsive management was asked about why this specific error occurs in the 'crab retained' and 'day' fields, they indicated that both of these fields are formatted as numeric values without any constraints on entering decimal values. It is likely that survey participants entered typos or non-sensical values while taking the survey.

4: Handling records reported as greater than 10 crab retained

In the dataset, if nmcrboriginal is greater than 10, RM changed these records to 10. In the 2022 data, there are 45 trips reporting more than 10 crab. It is likely that these records are people reporting an estimate of their entire catch for the season in a single fishing event. It is also possible that these records really are all caught in a single fishing event. For several of these records, other

fishing events are recorded for that same fisher, indicating that these are not the sum of the catch for the entire season. It is also possible both scenarios could be occurring. Regardless of the intent of the respondent, these reported crabs should be accounted for. In summer 2022, the majority (39 of 45) of these catches were recorded in Puget Sound marine areas with the remaining 6 recorded as either as 'unknown' or 'other marine area'. This issue did not occur in the 2023 data provided by Responsive Management. Any trips with catch greater than a daily limit were retained as shown in the nmcrb original, except the trips with unknown catch, which were treated separately (see #7 below).

Example: Non-respondent Vcid 114398711 harvested 11 crabs (nmcrboriginal) during a trip but nmcrb was converted to 10.

Vcid	month	whtdy	wchma	nmcrb	nmcrboriginal
114398711	2	32.000	13	10	11
114400184	5	32.000	3	10	32
114413380	4	32.000	6	10	12
114413380	2	32.000	6	6	6

Reccomendation: Do not modify these records and keep all crab greater than 10 as their original value.

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt these recommendations for processing the 2022 and 2023 Responsive Management telephone/email survey data.

5: Handling crab coded as '98':

If the crabber being interviewed did not recall the number of crabs harvested for a given trip, their catch would be coded as 99 (unknown # of crab). There are 5 trips that report the number of crabs harvested (nmcrboriginal) for a single trip as 98. RM has changed these records to 99 because they assumed they were entry errors when survey participants were attempting to enter 99. Besides these 98 records, the maximum crab recorded on a single trip is 50. Given that 98 is so close to 99, and that there are no trips recording crab between 50 and 98, it is likely that these 98 records are intended to be 99 (unknown # of crab) and are mistakenly entered as 98. Another possibility is that these fishers put in 98 intentionally as the maximum crab you could enter either because they actually retained that many crabs (or more), or to intentionally provide false data. This is possible as all 98 records are from the online survey (compared to phone). You would think that if any person wanted to intentionally and falsely report the maximum amount of crab, these false reports would be weeded out by the RM employee conducting the interview over the phone. In the 2022 data, 3 unique fishers entered 98 crab for a given trip, with one of these fishers entering 98 crab for 3 different trips. There were two 98 entries in the 2023 data by two separate crabbers.

Currently, Responsive Management treats these 98 responses as 99 (unknown # of crab). When asked about the presence of these 98 records, RM indicated that records shown as 98 for the number of crab harvested are indeed data entry errors and should be converted to 99 (unknown number for crab harvested). When asked about how survey participants were asked to enter crab retained for each trip into the survey itself, they responded, *‘Respondents were asked to key in their response manually, and could enter anything between 1 and 99, with 99 designated as don't know’*. These online entry methods are analogous to what was entered for those who were interviewed over the phone.

Example: For non-respondent Vcid ‘114461583’; Trip 1) The # of crab is set to 99; Trip 2) The # of crab is set to 98. See table below:

Vcid	month	whtdy	wchma	nmcra	nmcraoriginal
114458252	2	32	4	3	98
114458252	3	32	4	3	98
114458252	3	32	4	3	98
114458252	3	32	4	5	5
114458252	3	32	4	5	5
114458252	4	32	4	5	5
114458252	4	32	4	5	5
114461583	2	32	3	3	99
114461583	4	32	3	3	98
114414702	2	32	4	3	98

Possible data processing approaches:

- 3) Convert each 98 as a 99 (unknown # of crab). This assumes these records are 99 mistakenly entered in as 98.
- 3) Keep 98 records as 98. This assumes these 98 entries actually represent 98 crab retained or more if 98 is the maximum crab respondents could enter.
- 3) Remove these records from the dataset entirely, assuming these records were entered in to intentionally provide false data.

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt method 2 (highlighted) for processing the 2022 and 2023 Responsive Management telephone/email survey data. The group agreed to reconsider adopting method 1 once more information is provided from Responsive Management about the survey tool and the data entry process. There was no support for method 3 among the TWG members.

On 12/12/2024 we received confirmation from Responsive Management that the survey respondents were asked to key in numerical values for catch. Values between 1 and 99 could be entered, with 99 meaning unknown catch. Given that the respondents were asked to manually key in the catch, it is most likely that the 98 entries represent key-in errors for what should be 99 (unknown). Given these findings and the structure of the survey, data processing method 1 was adopted for the 98 entries for the number of crab (method 1).

6: Handling crab coded as 32:

If the crabber being interviewed did not recall the day (whtdy) of harvest for a given trip, the day would be coded as 32 (unknown day). In the 2022 data, there are 3 cases that report the number of **crabs** harvested (nmcrboriginal) as 32. It is possible that this actually corresponds to an unknown amount of crab mistakenly given the code for unknown day. Of these 3 records, 2 of them responded to the survey via email and 1 responded via phone. In some cases, other trips by these same fishers are set to 99 (unknown # of crab). In the 2023 data there were 6 trips where both the day and number crab retained were entered as 32.

Example: For non-respondent Vcid '114400184'; Trip 1) The # of crabs is set to 99 and the day is set to 32; Trip 2) The # of crab is set to 32 and the day is set to 32. See table below:

Vcid	month	whtdy	wchma	nmcrb	nmcrboriginal
114400184	5	32	3	3	99
114400184	5	32	3	10	32
114417322	2	32	7	3	99
114417322	3	32	7	3	99
114417322	3	32	7	3	99
114417322	2	32	7	10	32
114463317	2	32	5	10	32
114463317	3	32	5	5	5
114463317	2	32	5	5	5
114463317	2	32	5	5	5
114463317	2	32	5	5	5
114463317	2	32	5	5	5
114463317	3	32	5	3	3

Possible data processing approaches:

- 2) Keep these records as 32 crab. This assumes these records are people accurately reporting 32 crab.
- 2) Change these 32 records to 99 (unknown # of crab). This assumes these records are people reporting an unknown number of crabs using the unknown code for days rather than the unknown code for crab.

On 12/09/2024, the Joint Shellfish Technical Working Group agreed to adopt method 1 (highlighted) for processing the 2022 and 2023 Responsive Management telephone/email survey data as it is the more conservative approach and not enough information is provided to support method 2 at this time.

7: Handling records of 99 (unknown # of crab retained):

Of the total non-respondents that caught and retained crab, ~20% could not recall how many crabs were retained on at least 1 of their trips. Currently, RM changes these values to the median value (3) of all respondents who did recall their crab retained. The TWG should settle on a better method to estimate unknown # of crab from non-respondents.

Vcid	month	whtdy	wchma	nmcrcb	nmcrcboriginal
114374900	3	32	13	3	99
114376211	2	32	4	3	99
114376211	5	32	13	3	99
114377350	2	32	5	3	99

Possible data processing approaches:

- 3) Replace all 99's (unknown # of crab) with a mean crab per trip for all respondents that recalled their crab retained, but not include those who retained more than 10 crab. This would assume that any crab retained over 10 are actually the full-season catch of crab reported on a single trip. Including these crabs would therefore bias the mean of each individual trip upward by including those who reported their entire season on a single trip (see #4 above). This also assumes the records with 98 crab retained are changed to 99 (unknown # of crab) (see #5 above).
- 3) Replace all 99's (unknown # of crab) with a mean crab per trip for all respondents that recalled their crab retained, **without** reducing those that retained more than 10 crab to 10 crab. This would assume that any crab retained over 10 are the accurately reported catch for a single trip (see #4 above). This also assumes the records with 98 crab retained are changed to 99 (unknown # of crab) (see #5 above).
- 3) Replace all 99's (unknown # of crab) with a mean crab per trip for all respondents that recalled their crab retained, **without** reducing those that retained more than 10 crab to 10 crab (see #4 above) and **without** changing the crab reported as 98 to 99 (see #5 above). This would assume that any crab retained over 10 and as 98 are the accurately reported catch for a single trip.

*Following some discussion about the proposed methods for handling the '99' entries during the 12/09/204 meeting, the Joint Shellfish Technical Working Group agreed to adopt a modified version of method 3 for processing the 2022 and 2023 Responsive Management telephone/email survey data. The agreed-to method retains the '98' catch entries and applies the **median** for imputing catch for the '99' entries, which is the most appropriate given the skewed distribution of the data. The group agreed to reconsider changing the '98' to '99' once more information is provided from Responsive Management about the survey tool and the data entry process.*

2022/2023 Data processing recommendation for handling catch for trips with unknown # of crab retained.

Replace all 99's (unknown # of crab) with a median crab per trip for all respondents that recalled their crab retained, **without** reducing those that retained more than 10 crab to 10 crab (see #4 above) and **without** changing the crab reported as 98 to 99 (see #5 above). This would assume that any crab retained over 10 and as 98 are the accurately reported catch for a single trip.

Action Item for future data processing considerations:

- Ask Responsive Management to provide access to the original online survey so TWG members to better understand how data was entered into the survey by the participant.

2026 update - considerations for future data processing of non-respondent trip data.

When processing rules were first being derived for the 2022 and 2023 summer and winter non-respondent survey data, the analysts began with a review of the available data fields acquired from the interviews. Survey data provided for 2022 included responses to a question that asked crabbers about the number of crabbing trips they had taken that season. The exact question and response appeared to depend on whether the crabber was reading directly off a catch record card or not, but one variation of this question was: "How many trips were taken during which you caught and kept Dungeness crab in Puget Sound? Preliminary review of the data indicated that the number of trips reported when asked directly (i.e., with the question above) did not always match up with the number of trip records for which that same crabber provided some form of date and catch information (which translates to the number of rows in the data frame that were associated with each crabber). At the time when the above data processing rules were developed, the analysts choose to use the number of trip records for which a crabber provided some form of date or catch information, v. the number of trips the crabber cited when asked directly. This decision was made for the following reasons:

- 1) The question "How many trips were taken during which you caught and kept Dungeness crab in Puget Sound?" was only asked in 2022, and therefore could not be used to apply a consistent approach to both survey years.

On further exploration of the non-respondent data set, it was determined that the trips question above may hold value in producing a more reliable and accurate estimate for the number of trips taken by non-respondent crabbers. It is recommended that future data processing of the 2022 non-respondent survey data include expanded comparison of the discrepancies between these two methods of identifying the number of trips taken by interviewed crabbers.

APPENDIX 5 – TRP Data Discrepancy Background

WDFW data was used to inform the Technical Review Panel which finalized its report on unreported Dungeness crab catch in 2021. The data came from recreational crabber interviews that were conducted during the years 2010 through 2016.

The 2010 - 2013 dataset contained information from 3,777 crabbers. When data was combined for TRP review, it was noted that for some records information in the comments did not agree with the field “Total number of crab recorded” in the 2010-2013 data file. The errors were observed in the data from 2011, 2012, and 2013 specifically. The impact of these are detailed in the table below.

Years	Crabbers	Dungeness Kept	Recorded Crab (TRP file)	Recorded Crab (Original data file)	TRP File % Unrecorded	Original File % Unrecorded
2010-2013	3,777	13,127	11,902	11,769	9.33%	10.35%
2015-2016*	1,456	6,088	5,764	5,764	5.32%	5.32%
Overall	5,233	19,215	17,666	17,533	8.06%	8.75%

**Data from interviews conducted from August 26, 2016 through August 30, 2016 were not included in the summary. The WILD Licensing system was compromised in this window and recreational crabbers were not required to record on CRCs.*

The 2015-2016 dataset contained information from 1,438 crabbers who were interviewed on dates where they were required to record their catch on CRCs. Crabbers interviewed from August 26, 2016 through August 30, 2016 were not included because they were not required to record on CRCs during this time span when the WILD Licensing system was compromised ([WDFW offers free fishing while license sales are suspended | Washington Department of Fish & Wildlife](#)). Catch for this week without the CRC requirement were accounted by substituting catch from the week prior with a 10% inflation. This resulted in overall increase in the 2016 August harvest by 7% when compared to the year prior. Crabbers interviewed in 2016 after August 30th were again required to record their Dungeness catch on CRCs ([WDFW resumes sale of fishing and hunting licenses | Washington Department of Fish & Wildlife](#)) and that data is included in the summary table below.

APPENDIX 6 – Ongoing and Future Work

The primary sections of the *Technical Working Group Recommendations for Catch Record Card -Based Puget Sound Recreational Harvest Estimation* document detail recommendations following the focused work of the TWG on CRC based recreational Dungeness catch estimation methods. Throughout its collaborative work on the topics detailed above, members also raised other important questions and concerns that were both directly related to recreational catch estimation and other components of the fishery. In keeping with the TWG's collaborative and objective charge mission, those topics are documented here without any prioritization for policy consideration for the TWG to focus on in future work. This list of items is intended to reflect topics and concerns raised by any member of the TWG and does not represent the opinions or ideas of the entire group. This should not be construed as a comprehensive list of all potential priorities that the TWG could focus on, nor should it be construed that members are in agreement that these potential topics should be focused on by the TWG.

Recreational Catch Estimation: Several topics related to recreational Dungeness crab catch estimation methodology, processes, and procedures could benefit from the directed focus of the TWG. Due to the complexity and scale of recreational catch estimation and the similarity to the topics focused on in this document, future work considerations related to recreational catch estimation have been consolidated into the following items which could be focused on individually or in aggregate by the TWG.

- **Regular Review and Assessment of Existing Catch Estimation/Accounting Methods:** The TWG could periodically review the catch estimation methods including when data streams are added, subtracted, or altered to ensure best practices for establishing a catch estimate are current and continue to use the best available methods and data. The TWG could also review the implementation status of TWG recommendations detailed in this report.
- **Charter Catch:** The methods for incorporating charter catch, “for-hire fishing” activity in general, and the total contribution of charter harvest to the recreational catch estimate remains unclear to the TWG. The TWG could document and review charter catch methods, charter participation, and documenting charter harvest could be incorporated in a follow-up effort of the TWG.
- **CRC-based Estimate Improvements and Validation:** With electronic-CRCs (e-CRCs) being implemented in the near future, the TWG could develop and evaluate methods for generating a recreational catch estimate using real-time data instead of post-season reporting and address other e-CRC methodological related topics.
- **Alternative Catch Estimation Methods:** The TWG could evaluate the efficacy and utility of catch estimation methods that do not rely on self-reported CRC data.
- **Alternative and Supplemental Survey Methods:** Survey data from several different survey types are used to generate recreational catch estimates. The TWG

could evaluate and design the surveys used to generate recreational catch estimates including:

- o Currently implemented creel and non-respondent surveys.
- o Proposed on-the-water creel surveys and flight-based effort estimates.
- **Integration of Additional Parameters into Recreational Crab Catch Estimation (i.e. crab weights):** The TWG could evaluate methods for integrating biometric or other parameters, such as crab weights, into recreational crab catch estimation methods.

Evaluating and Updating Methodology to Estimate the Proportion of Unreported Catch in Commercial and C&S fisheries: The TWG could develop methods for evaluating non-recreational EUC's captured below.

- **State / Treaty Commercial EUC:** The proportion of State and Treaty commercial catch that contributes to the EUC is assumed to be 1%. The TWG could evaluate data to inform updating the commercial EUC. **Treaty C&S EUC:** The proportion of treaty C&S catches that contributes to the EUC is assumed to be 6%. The TWG could evaluate data to inform updating the C&S EUC.

Evaluating Indirect impacts on Dungeness Crab due to Fishery Induced Handling and Gear-loss Impacts: Working towards an estimate of total fishery induced mortality is a concern to co-managers. The TWG could start by evaluate the following indirect fishery induced sources of mortality.

- **Fishery Specific Rates of Handling Mortality:** The TWG could assess and develop estimates for fishery induced mortality rates for Treaty Commercial, State Commercial, State Recreational, and Treaty C&S fisheries independently.
- **Derelict Gear / Ghost Fishing:** The TWG could develop methods to estimate rates of gear loss for each fishery type. The TWG could also develop methods to estimate the mortality from lost gear for each fishery type.